# Performance-Constrained Adaptive Sliding Mode Control for Guaranteed Soft Precise Landing Using Optic Flow

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***Abstract*—This paper proposes a *scale-independent* adaptive control strategy that allows a quadrotor to rely solely on image output to achieve a soft precise landing on mobile platforms. This scheme proposes an adaptive sliding mode control strategy with theoretical stability guarantees that incorporates performance and input constraints to achieve desired transient and steady-state performance in the presence of uncertainty arising from wind gusts, ground effects, and possible actuator saturation. In particular, detailed Lyapunov stability analysis is used to demonstrate that the optic flow error converges exponentially to a small prescribed bound in steady-state, which is shown to be the key to achieving a guaranteed soft touchdown. Extensive experimental results substantiate the efficacy of the proposed strategy using a micro-quadrotor platform in a controlled environment. Moreover, a laterally stabilized soft landing with tailored descent behavior is also shown using a mini-quadrotor platform in real-world (unstructured) settings in the presence of wind gusts. Performance comparison studies with an image-based fixed-gain proportional-integral control strategy and an pose-based control strategy are also included, demonstrating the superior performance of the proposed strategy in comparison with these alternative designs.**

Manuscript received XXXXX 00, 0000; revised XXXXX 00, 0000; accepted XXXXX 00, 0000.
This work was supported in part by the SERB Core Research Grant CRG/2021/006872 and the Space Technology Cell Grant STC/P-486.
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The experimental results can be found in this video: https://youtu.be/1wcuQ9zRay0.

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***Index Terms*—Optic flow regulation, adaptive sliding mode control, guaranteed precise landing, visual guidance, prescribed performance constraint**

## Nomenclature

| | | |
|---|---|---|
| $t$ | = | time, s |
| $m$ | = | mass, kg |
| $\boldsymbol{r}$ | = | radial position vector, m |
| $\boldsymbol{v}$ | = | linear velocity vector, m/s |
| $\boldsymbol{\omega}$ | = | angular velocity vector, rad/s |
| $\boldsymbol{F}$ | = | force vector, N |
| $J$ | = | moment of inertia matrix, kg m$^2$ |
| $\boldsymbol{\tau}$ | = | torque vector, N m |
| $T$ | = | total thrust of the rotors, N |
| $\hat{\boldsymbol{r}}$ | = | image point |
| $\boldsymbol{s}$ | = | normalized position vector, rad |
| $\alpha$ | = | image orientation, rad |
| $\boldsymbol{h}$ | = | optical flow, rad/s |
| $\boldsymbol{q}$ | = | system output vector |
| $\boldsymbol{u}$ | = | system input vector |
| $\beta$ | = | uncertainty parameter |
| $\boldsymbol{c}$ | = | vector representing known system dynamics |
| $\boldsymbol{d}$ | = | vector representing unknown system dynamics |
| $\boldsymbol{p}$ | = | performance function vector |
| $V$ | = | Lyapunov function |

## I. INTRODUCTION

UNMANNNED Aerial Vehicles (UAVs) have emerged as indispensable assets across diverse mission profiles, ranging from defense and disaster relief to logistics and scientific exploration. A critical capability underpinning these applications is the ability to execute soft, precise landings on designated targets, which may themselves be mobile platforms such as ships, vehicles, or ground robots. The need for such landings arises from the dual imperatives of mission reliability and system safety: imprecise or hard landings risk damage to the UAV, payload, or target infrastructure, thereby compromising operational success and increasing maintenance costs.

Autonomous landing of UAVs has been extensively studied, with solutions broadly categorized into position-based and image-based visual servoing (PBVS and IBVS) frameworks. PBVS approaches rely on explicit 3D pose reconstruction using metric scale information obtained from sensors such as LiDAR, stereo cameras, or GPS [1], [2]. While these methods provide accurate pose estimates under ideal conditions, their performance degrades significantly in GPS-denied environments and in the presence of sensor noise, calibration errors, or scale drift.

In contrast, Image-Based Visual Servoing (IBVS) methods directly utilize image features to synthesize control inputs, thereby eliminating the need for explicit 3D reconstruction [3], [4]. Foundational works have established the IBVS framework using the interaction matrix to relate image feature dynamics to camera motion [3]. Extensions such as moment-based visual servoing further improve robustness to feature extraction and tracking

uncertainties [5]. However, classical IBVS approaches are primarily developed for robotic manipulators and kinematic systems, and their direct application to UAVs introduces significant challenges due to highly nonlinear dynamics and underactuation.

Several studies have extended IBVS to aerial platforms [6], [7], [8], [9]. These works incorporate UAV dynamics into the visual servoing framework and demonstrate successful landing and tracking performance. Nevertheless, they often rely on implicit or explicit scale information and assume ideal sensing conditions. Moreover, the control designs in these approaches typically lack robustness against external disturbances, actuator saturation, and model uncertainties, which are critical during the terminal landing phase.

More recent works have explored adaptive and learning-based approaches to enhance landing performance [10], [11], [12], [13]. These methods aim to improve robustness and adaptability under uncertain environments. However, they generally depend on computationally intensive perception pipelines or learning models and often lack rigorous theoretical guarantees for stability and convergence. In particular, the presence of actuator constraints, ground effects, and wind disturbances is either neglected or only partially addressed.

A fundamental limitation across existing visual servoing approaches is the dependence on metric scale or depth estimation, which is inherently ambiguous in monocular vision systems. While some works attempt to mitigate this issue through sensor fusion or estimation techniques, these solutions introduce additional complexity and are sensitive to noise and calibration errors. Furthermore, most existing methods do not explicitly enforce performance constraints required for soft landing, such as bounded velocity near touchdown and avoidance of rebound.

In summary, despite significant progress in visual servoing-based UAV landing, there remains a lack of control strategies that simultaneously achieve scale independence, robustness to uncertainties and disturbances, and rigorous stability guarantees under actuator constraints (refer *Table* I). This gap motivates the development of the proposed performance-constrained adaptive control framework.

The main contributions of this paper are summarized as follows:

(a) Unlike conventional IBVS and PBVS approaches that rely on metric scale or depth estimation, the proposed method leverages optic flow as the primary feedback variable. Since optic flow inherently encodes the ratio of velocity to distance, it eliminates the need for explicit scale recovery, thereby avoiding issues related to scale ambiguity, calibration errors, and sensor noise.

(b) The proposed Performance-constrained Leakage-type Adaptive Sliding Mode Control (PLASMC) framework explicitly accounts for system uncertainties, including exogenous disturbances (e.g., wind gusts), ground effects, actuator faults, and input saturation. In contrast to existing IBVS-based UAV controllers, which are predominantly designed under idealized assumptions, the proposed approach incorporates these practical nonlinearities directly into the control design.

(c) The control law is synthesized within a Lyapunov-based framework that guarantees global exponential convergence of the optic flow error to a predefined small neighborhood around the origin. The integration of performance constraints ensures that the transient and steady-state behavior of the system satisfies the functional requirements of soft landing, including bounded descent velocity and smooth touchdown.

(d) The adaptive mechanism does not require prior knowledge of the upper bounds of system uncertainties, thereby enhancing practical deployability. Furthermore, the leakage-type adaptation mitigates parameter drift and ensures boundedness of all closed-loop signals.

Collectively, these features enable the proposed approach to achieve robust, scale-free, and constraint-aware control for soft landing, which is not simultaneously addressed in existing literature.

The rest of the paper is structured as follows. Section II presents the mathematical description of the optic flow-guided soft landing challenge for quadrotor UAVs, and Section III outlines the control design and associated stability analysis for regulation of optical flow induced by vertical motion. The details of the experimental setup and the experimental results, validating the performance and robustness of the proposed control strategy, are provided in Section **??**. Finally, Section V presents the conclusions of this study.

## II. PRELIMINARIES & PROBLEM FORMULATION

This section outlines the system dynamics and control strategy of an underactuated quadrotor UAV platform using scale-independent target parameters and formulates the problem to address the challenges in vision-based precise landing.

### A. Virtual Feature Points

Consider a camera frame $\mathcal{C} = \{O_\mathrm{c}\ X_\mathrm{c}\ Y_\mathrm{c}\ Z_\mathrm{c}\}$ rigidly attached to a monocular camera such that its $X_\mathrm{c}$ and $Y_\mathrm{c}$ axes are aligned with the image horizontal and vertical directions respectively and its $Z_\mathrm{c}$ axis is aligned with the optical axis of the camera, pointing towards the scene. Assume that the camera captures a feature point ${}^{\mathcal{C}}\hat{\boldsymbol{r}}$ in the image plane, which can be represented using the common pinhole camera model as:

$$ {}^{\mathcal{C}}\hat{\boldsymbol{r}} = \begin{bmatrix} {}^{\mathcal{C}}\hat{x} \\ {}^{\mathcal{C}}\hat{y} \end{bmatrix} = \frac{1}{{}^{\mathcal{C}}z} \begin{bmatrix} {}^{\mathcal{C}}x \\ {}^{\mathcal{C}}y \end{bmatrix}, \tag{1} $$

TABLE I: Comparison of Visual Servoing-Based UAV Landing Approaches

| Reference | Scale-Free | Robustness | Disturbance Handling | Actuator Constraints | Stability Guarantee |
|---|---|---|---|---|---|
| Jabbari [6] | × | × | × | × | × |
| Xie [8], [9], [7] | × | × | × | × | × |
| Salehi [1] | × | ✓ | ✓ | × | × |
| Cho [2] | × | ✓ | ✓ | × | × |
| Chen [10] | × | ✓ | ✓ | × | × |
| Bouazza [11] | × | ✓ | ✓ | × | × |
| Kamath [12] | × | ✓ | ✓ | × | × |
| Zhang [13] | × | ✓ | ✓ | × | × |
| **Proposed** | ✓ | ✓ | ✓ | ✓ | ✓ |

where the feature point ${}^{\mathcal{C}}\hat{\boldsymbol{r}}$ is the perspective projection of a 3-D point in the camera frame ${}^{\mathcal{C}}\boldsymbol{r} = [{}^{\mathcal{C}}x,\, {}^{\mathcal{C}}y,\, {}^{\mathcal{C}}z]^\top$ on the image plane.

Now consider a virtual camera frame $\mathcal{V} = \{O_\mathrm{v}\ X_\mathrm{v}\ Y_\mathrm{v}\ Z_\mathrm{v}\}$, which moves with the camera frame $\mathcal{C}$ but in such a way that the $X_\mathrm{v}$–$Y_\mathrm{v}$ plane is always horizontal, i.e., parallel to the $X_\mathrm{i}$–$Y_\mathrm{i}$ plane. Associated with $\mathcal{V}$, let us also define a virtual image plane, whose position and orientation w.r.t. $\mathcal{V}$ is the same as the actual image plane w.r.t. $\mathcal{C}$. The virtual image plane transformation is conducted by using the roll $\phi$ and pitch $\theta$ angles of the UAV. The 3-D point in the camera frame ${}^{\mathcal{C}}\boldsymbol{r}$ is expressed in the virtual coordinate frame as ${}^{\mathcal{V}}\boldsymbol{r} = [{}^{\mathcal{V}}x,\, {}^{\mathcal{V}}y,\, {}^{\mathcal{V}}z]^\top$. The relationship between ${}^{\mathcal{C}}\boldsymbol{r}$ and ${}^{\mathcal{V}}\boldsymbol{r}$ can be expressed as:

$$ {}^{\mathcal{V}}\boldsymbol{r} = {}^{\mathcal{V}}R_{\mathcal{C}}\, {}^{\mathcal{C}}\boldsymbol{r}, \tag{2} $$

where the rotation matrix ${}^{\mathcal{V}}R_{\mathcal{C}} = R_y(\theta)R_x(\phi)$ transforms vectors from camera frame $\mathcal{C}$ to the virtual camera frame $\mathcal{V}$. Using (2) and the pinhole camera model (1), virtual feature point ${}^{\mathcal{V}}\hat{\boldsymbol{r}}$ is computed in the virtual image plane as:

$$ {}^{\mathcal{V}}\hat{\boldsymbol{r}} = \begin{bmatrix} {}^{\mathcal{V}}\hat{x} \\ {}^{\mathcal{V}}\hat{y} \end{bmatrix} = \frac{1}{{}^{\mathcal{V}}z}\begin{bmatrix} {}^{\mathcal{V}}x \\ {}^{\mathcal{V}}y \end{bmatrix} = \begin{bmatrix} \frac{C_\theta\, {}^{\mathcal{C}}\hat{x} + S_\theta S_\phi\, {}^{\mathcal{C}}\hat{y} + S_\theta C_\phi}{-S_\theta\, {}^{\mathcal{C}}\hat{x} + C_\theta S_\phi\, {}^{\mathcal{C}}\hat{y} + C_\theta C_\phi} \\ \frac{C_\theta\, {}^{\mathcal{C}}\hat{y} - S_\phi}{-S_\theta\, {}^{\mathcal{C}}\hat{x} + C_\theta S_\phi\, {}^{\mathcal{C}}\hat{y} + C_\theta C_\phi} \end{bmatrix} \tag{3} $$

## B. Dynamics of Quadrotor-Target System

Achieving soft precise landing on a moving target requires a thorough understanding of the system dynamics and control architecture of an underactuated quadrotor UAV platform. First, consider a body-fixed frame $\mathcal{B} = \{O_\mathrm{b}\ X_\mathrm{b}\ Y_\mathrm{b}\ Z_\mathrm{b}\}$ rigidly attached to the UAV in forward-right-down (FRD) configuration. Let the linear and angular velocities of the UAV be denoted as ${}^{\mathcal{B}}\boldsymbol{v}_\mathrm{b}$ and ${}^{\mathcal{B}}\boldsymbol{\omega}_\mathrm{b}$ respectively. Now, using the Newton-Euler formulation, the UAV flight dynamics can be expressed as follows:

$$ \begin{bmatrix} {}^{\mathcal{B}}\dot{\boldsymbol{v}}_\mathrm{b} \\ {}^{\mathcal{B}}\dot{\boldsymbol{\omega}}_\mathrm{b} \end{bmatrix} = \begin{bmatrix} B_v & O_{3\times3} \\ O_{3\times3} & B_\omega \end{bmatrix} \begin{bmatrix} {}^{\mathcal{B}}\boldsymbol{F}_u \\ {}^{\mathcal{B}}\boldsymbol{\tau}_u \end{bmatrix} + \begin{bmatrix} \boldsymbol{c}_v \\ \boldsymbol{c}_\omega \end{bmatrix} + \begin{bmatrix} {}^{\mathcal{B}}\boldsymbol{F}_\mathrm{d} \\ {}^{\mathcal{B}}\boldsymbol{\tau}_\mathrm{d} \end{bmatrix} \tag{4} $$

where

$$ \begin{aligned} B_v &= \frac{1}{m} I_{3\times3}, & \boldsymbol{c}_v &= \frac{1}{m}\, {}^{\mathcal{B}}\boldsymbol{F}_\mathrm{g} - {}^{\mathcal{B}}\boldsymbol{\omega}_\mathrm{b} \times {}^{\mathcal{B}}\boldsymbol{v}_\mathrm{b} \\ B_\omega &= J^{-1}, & \boldsymbol{c}_\omega &= -J^{-1}\left({}^{\mathcal{B}}\boldsymbol{\omega}_\mathrm{b} \times (J\, {}^{\mathcal{B}}\boldsymbol{\omega}_\mathrm{b})\right) \end{aligned} $$

Here $m$ is the body mass, and $J$ is the body inertia matrix. The control force ${}^{\mathcal{B}}\boldsymbol{F}_u$ is defined as ${}^{\mathcal{B}}\boldsymbol{F}_u = \begin{bmatrix} 0 & 0 & {}^{\mathcal{B}}T_u \end{bmatrix}^\top$ with ${}^{\mathcal{B}}T_u$ being the total rotor thrust. ${}^{\mathcal{B}}\boldsymbol{F}_\mathrm{g}$ is the gravitational force ${}^{\mathcal{B}}\boldsymbol{F}_\mathrm{g}$ and ${}^{\mathcal{B}}\boldsymbol{\tau}_u$ is the control torque. The unknown bounded disturbances are denoted by ${}^{\mathcal{B}}\boldsymbol{F}_\mathrm{d}$ and ${}^{\mathcal{B}}\boldsymbol{\tau}_\mathrm{d}$.

Next, consider a downward facing monocular camera mounted onboard the UAV platform. Assuming the optical axis of the camera is aligned with the $Z_\mathrm{b}$ axis of body frame $\mathcal{B}$, the camera continuously captures the target. This study utilizes the target image dynamics as given below:

$$ \dot{\boldsymbol{h}} = B_h\, {}^{\mathcal{V}}\boldsymbol{F}_u + \boldsymbol{c}_h + \boldsymbol{d}_h, \tag{5} $$

where

$$ \begin{aligned} B_h &= -\frac{1}{m\, {}^{\mathcal{V}}z_\mathrm{t}} I_{3\times3}, \\ \boldsymbol{c}_h &= -{}^{\mathcal{V}}\dot{\boldsymbol{\omega}}_\mathrm{v} \times \boldsymbol{s} - {}^{\mathcal{V}}\boldsymbol{\omega}_\mathrm{v} \times \left({}^{\mathcal{V}}\boldsymbol{\omega}_\mathrm{v} \times \boldsymbol{s}\right) - 2\left({}^{\mathcal{V}}\boldsymbol{\omega}_\mathrm{v} \times \boldsymbol{h}\right) \\ &\quad - \left[\left(\boldsymbol{h} - {}^{\mathcal{V}}\boldsymbol{\omega}_\mathrm{v} \times \boldsymbol{s}\right) \cdot \hat{\boldsymbol{e}}_3\right] \boldsymbol{h} - \dot{\boldsymbol{h}}_\mathrm{d} \\ \boldsymbol{d}_h &= \frac{{}^{\mathcal{V}}\boldsymbol{a}_\mathrm{t}}{{}^{\mathcal{V}}z_\mathrm{t}} - \left(\frac{{}^{\mathcal{V}}\boldsymbol{F}_\mathrm{g} + {}^{\mathcal{V}}\boldsymbol{F}_\mathrm{d}}{m\, {}^{\mathcal{V}}z_\mathrm{t}} - {}^{\mathcal{V}}\boldsymbol{\omega}_\mathrm{b} \times \frac{{}^{\mathcal{V}}\boldsymbol{v}_\mathrm{b}}{{}^{\mathcal{V}}z_\mathrm{t}}\right). \end{aligned} $$

Here $\boldsymbol{h} = {}^{\mathcal{V}}\boldsymbol{v}_{\mathrm{t/v}}/{}^{\mathcal{V}}z_\mathrm{t}$ is the optical flow representing visual velocity of the target and ${}^{\mathcal{V}}\boldsymbol{\omega}_\mathrm{v} = [0, 0, -\dot{\psi}]^\top$ is the angular rate of the virtual plane in the virtual frame $\mathcal{V}$.

## C. Pose-based Flight Controller

In order to achieve target tracking using a pose-based flight controller, the overall control architecture is a cascaded PID controller with outer-loop operating in the inertial frame $\mathcal{I} = \{O_\mathrm{i}\ X_\mathrm{i}\ Y_\mathrm{i}\ Z_\mathrm{i}\}$ and the inner-loop in the body frame $\mathcal{B}$. The outer-loop controller is designed to generate desired velocity ${}^{\mathcal{I}}\boldsymbol{v}_\mathrm{d}$ command that drives the UAV from the instantaneous position ${}^{\mathcal{I}}\boldsymbol{r}_\mathrm{b}$ to the known target position ${}^{\mathcal{I}}\boldsymbol{r}_\mathrm{t}$. Then, it generates desired acceleration ${}^{\mathcal{I}}\boldsymbol{a}_\mathrm{d}$ command that drives the instantaneous UAV velocity ${}^{\mathcal{I}}\boldsymbol{v}_\mathrm{b}$ to the desired velocity ${}^{\mathcal{I}}\boldsymbol{v}_\mathrm{d}$. The control laws are shown below:

$$ \begin{aligned} {}^{\mathcal{I}}\boldsymbol{v}_\mathrm{d} &= -K_{rp}\, {}^{\mathcal{I}}\boldsymbol{r}_\mathrm{e} - K_{ri}\int {}^{\mathcal{I}}\boldsymbol{r}_\mathrm{e} dt - K_{rd}\frac{d\, {}^{\mathcal{I}}\boldsymbol{r}_\mathrm{e}}{dt} \\ {}^{\mathcal{I}}\boldsymbol{a}_\mathrm{d} &= K_{vp}\, {}^{\mathcal{I}}\boldsymbol{v}_\mathrm{e} - K_{vi}\int {}^{\mathcal{I}}\boldsymbol{v}_\mathrm{e} dt - K_{vd}\frac{d\, {}^{\mathcal{I}}\boldsymbol{v}_\mathrm{e}}{dt}, \end{aligned} \tag{6} $$

where ${}^{\mathcal{I}}\boldsymbol{r}_\mathrm{e} = {}^{\mathcal{I}}\boldsymbol{r}_\mathrm{b} - {}^{\mathcal{I}}\boldsymbol{r}_\mathrm{t}$ represents the relative position between the UAV and the target and ${}^{\mathcal{I}}\boldsymbol{v}_\mathrm{e} = {}^{\mathcal{I}}\boldsymbol{v}_\mathrm{b} - {}^{\mathcal{I}}\boldsymbol{v}_\mathrm{d}$.

Since the centrifugal force ${}^{\mathcal{B}}\boldsymbol{\omega}_{\mathrm{b}} \times (m\,{}^{\mathcal{B}}\boldsymbol{v}_{\mathrm{b}})$, given in (4), is nullified in the inertia frame, the desired acceleration ${}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}$ is assumed to be achieved solely by the total rotor control thrust ${}^{\mathcal{B}}T_u$ and its direction, after compensating for gravity and disturbances. The relationship between the desired acceleration (${}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}$) and the thrust force applied along the desired body Z-axis (${}^{\mathcal{B}}T_u$) is derived as:

$$ {}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}} = \frac{1}{m}\,{}^{\mathcal{I}}R_{\mathcal{B}}\,{}^{\mathcal{B}}\boldsymbol{F}_u = \frac{1}{m}\,{}^{\mathcal{I}}R_{\mathcal{B}}\begin{bmatrix}0 & 0 & {}^{\mathcal{B}}T_u\end{bmatrix}^{\top}, \quad (7) $$

where the rotation matrix ${}^{\mathcal{I}}R_{\mathcal{B}}$ transforms vectors from body frame ($\mathcal{B}$) to the inertial frame ($\mathcal{I}$).

The desired roll ($\phi_{\mathrm{d}}$), pitch ($\theta_{\mathrm{d}}$), and total thrust (${}^{\mathcal{B}}T_u$) commands are then analytically derived by inverting the acceleration ${}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}$ in (7) using its components (${}^{\mathcal{I}}a_{\mathrm{d}_x}$, ${}^{\mathcal{I}}a_{\mathrm{d}_y}$, ${}^{\mathcal{I}}a_{\mathrm{d}_z}$):

$$ \begin{aligned} \theta_{\mathrm{d}} &= \tan^{-1}\left(\frac{-S_{\psi}\,{}^{\mathcal{I}}a_{\mathrm{d}_x} - C_{\psi}\,{}^{\mathcal{I}}a_{\mathrm{d}_y}}{{}^{\mathcal{I}}a_{\mathrm{d}_z}}\right) \\ \phi_{\mathrm{d}} &= \tan^{-1}\left(\frac{C_{\theta_{\mathrm{d}}}\left(C_{\psi}\,{}^{\mathcal{I}}a_{\mathrm{d}_x} - S_{\psi}\,{}^{\mathcal{I}}a_{\mathrm{d}_y}\right)}{{}^{\mathcal{I}}a_{\mathrm{d}_z}}\right) \\ {}^{\mathcal{B}}T_u &= -\frac{m\,{}^{\mathcal{I}}a_{\mathrm{d}_z}}{C_{\theta_{\mathrm{d}}}C_{\phi_{\mathrm{d}}}}, \end{aligned} \quad (8) $$

in which $S_x = sin(x)$ and $C_x = cos(x)$. Combining the derived attitude ($\phi_{\mathrm{d}}, \theta_{\mathrm{d}}$) with the known target heading $\psi_{\mathrm{t}}$, the desired attitude $\boldsymbol{\Phi}_{\mathrm{d}} = [\phi_{\mathrm{d}}, \theta_{\mathrm{d}}, \psi_{\mathrm{t}}]^{\top}$ and thrust (${}^{\mathcal{B}}T_u$) commands are then passed to the inner-loop controller to execute the required control effort.

The inner-loop controller uses a cascade of PID loops to regulate the attitude ($\boldsymbol{\Phi}_{\mathrm{b}}$) and angular body rates (${}^{\mathcal{B}}\boldsymbol{\omega}_{\mathrm{b}}$) of the UAV. The goal is to generate desired attitude rates ($\dot{\boldsymbol{\Phi}}_{\mathrm{d}}$) command that helps the UAV to accurately follow the desired attitude ($\boldsymbol{\Phi}_{\mathrm{d}}$) command generated by the outer-loop controller. The desired attitude rates ($\dot{\boldsymbol{\Phi}}_{\mathrm{d}}$) are transformed into the desired angular body rates (${}^{\mathcal{B}}\boldsymbol{\omega}_{\mathrm{d}}$) using the kinematic transformation matrix $W(\phi_{\mathrm{d}}, \theta_{\mathrm{d}})$:

$$ {}^{\mathcal{B}}\boldsymbol{\omega}_{\mathrm{d}} = W(\phi_{\mathrm{d}}, \theta_{\mathrm{d}})\dot{\boldsymbol{\Phi}}_{\mathrm{d}}, \quad (9) $$

where matrix $W(\phi_{\mathrm{d}}, \theta_{\mathrm{d}})$ relates the inertial Euler rates to the body frame angular rates, based on the desired roll ($\phi_{\mathrm{d}}$) and pitch ($\theta_{\mathrm{d}}$) angles as shown below:

$$ W(\phi_{\mathrm{d}}, \theta_{\mathrm{d}}) = \begin{bmatrix} 0 & C_{\phi_{\mathrm{d}}} & S_{\phi_{\mathrm{d}}}C_{\theta_{\mathrm{d}}} \\ 1 & 0 & -S_{\theta_{\mathrm{d}}} \\ 0 & S_{\phi_{\mathrm{d}}} & -C_{\phi_{\mathrm{d}}}C_{\theta_{\mathrm{d}}} \end{bmatrix} \quad (10) $$

Then, the inner-loop controller generates desired angular acceleration ${}^{\mathcal{B}}\dot{\boldsymbol{\omega}}_{\mathrm{d}}$ command that drives the current body rate ${}^{\mathcal{B}}\boldsymbol{\omega}_{\mathrm{b}}$ of the UAV to the desired body rate ${}^{\mathcal{B}}\boldsymbol{\omega}_{\mathrm{d}}$. The control laws are shown below:

$$ \begin{aligned} \dot{\boldsymbol{\Phi}}_{\mathrm{d}} &= -K_{\Phi p}\boldsymbol{\Phi}_{\mathrm{e}} - K_{\Phi i}\int \boldsymbol{\Phi}_{\mathrm{e}}dt - K_{\Phi d}\frac{d\boldsymbol{\Phi}_{\mathrm{e}}}{dt} \\ {}^{\mathcal{B}}\dot{\boldsymbol{\omega}}_{\mathrm{d}} &= -K_{\omega p}\,{}^{\mathcal{B}}\boldsymbol{\omega}_{\mathrm{e}} - K_{\omega i}\int {}^{\mathcal{B}}\boldsymbol{\omega}_{\mathrm{e}}dt - K_{\omega d}\frac{d\,{}^{\mathcal{B}}\boldsymbol{\omega}_{\mathrm{e}}}{dt}, \end{aligned} \quad (11) $$

where $\boldsymbol{\Phi}_{\mathrm{e}} = \boldsymbol{\Phi}_{\mathrm{b}} - \boldsymbol{\Phi}_{\mathrm{d}}$ and $\boldsymbol{\omega}_{\mathrm{e}} = {}^{\mathcal{B}}\boldsymbol{\omega}_{\mathrm{b}} - {}^{\mathcal{B}}\boldsymbol{\omega}_{\mathrm{d}}$. The desired angular acceleration ${}^{\mathcal{B}}\dot{\boldsymbol{\omega}}_{\mathrm{d}}$ is then used with the rigid body dynamics equation (4) to calculate the necessary desired control torque (${}^{\mathcal{B}}\boldsymbol{\tau}_u$):

$$ {}^{\mathcal{B}}\boldsymbol{\tau}_u = J\,{}^{\mathcal{B}}\dot{\boldsymbol{\omega}}_{\mathrm{d}} + {}^{\mathcal{B}}\boldsymbol{\omega}_{\mathrm{d}} \times (J\,{}^{\mathcal{B}}\boldsymbol{\omega}_{\mathrm{d}}) \quad (12) $$

The final commands sent to the motor mixing algorithm are the combined thrust-torque command $\boldsymbol{u} = [{}^{\mathcal{B}}T_u, {}^{\mathcal{B}}\boldsymbol{\tau}_u^{\top}]^{\top}$.

## D. Problem Statement

This study proposes an image-based flight controller that uses scale-independent target parameters, instead of scale-dependent state variables ${}^{\mathcal{I}}\boldsymbol{r}_{\mathrm{b}}$ and ${}^{\mathcal{I}}\boldsymbol{v}_{\mathrm{b}}$ given in (6), to maneuver the UAV towards the target without explicitly knowing the target position or heading. These target parameters are obtained using a downward-facing monocular camera mounted onboard the UAV platform. Using the common pinhole camera model (1), the normalized target position $\boldsymbol{s}$ can be obtained using the target image point ${}^{\mathcal{V}}\hat{\boldsymbol{r}}$ as shown below:

$$ \boldsymbol{s} = \begin{bmatrix} {}^{\mathcal{V}}\hat{\boldsymbol{r}} \\ 1 \end{bmatrix} = \frac{{}^{\mathcal{V}}\boldsymbol{r}_{\mathrm{t}}}{{}^{\mathcal{V}}z_{\mathrm{t}}}, \quad (13) $$

where $\boldsymbol{s}$ represents the perspective projection of the target point in the virtual frame ${}^{\mathcal{V}}\boldsymbol{r}_{\mathrm{t}} = [{}^{\mathcal{V}}x_{\mathrm{t}}, {}^{\mathcal{V}}y_{\mathrm{t}}, {}^{\mathcal{V}}z_{\mathrm{t}}]^{\top}$ on the virtual image plane. Since the onboard camera has limited capabilities, the target image point ${}^{\mathcal{V}}\hat{\boldsymbol{r}}$ is visible to it only when the target stays within its field of view. Hence, ${}^{\mathcal{V}}\hat{\boldsymbol{r}}$ is subjected to the following visibility constraints:

$$ -\frac{\mathcal{R}}{2f} \leq {}^{\mathcal{V}}\hat{\boldsymbol{r}} \leq \frac{\mathcal{R}}{2f}, \quad (14) $$

where $\mathcal{R} = [r_h, r_w]^{\top}$ represents image resolution in pixels.

Hence, the visual kinematics is given as:

$$ \dot{\boldsymbol{s}} = \boldsymbol{h} - {}^{\mathcal{V}}\boldsymbol{\omega}_{\mathrm{v}} \times \boldsymbol{s} - \left[\left(\boldsymbol{h} - {}^{\mathcal{V}}\boldsymbol{\omega}_{\mathrm{v}} \times \boldsymbol{s}\right) \cdot \hat{\boldsymbol{e}}_3\right]\boldsymbol{s}, \quad (15) $$

Given the desired normalized target position as $\boldsymbol{s}_{\mathrm{d}} = [\hat{\boldsymbol{r}}_{\mathrm{d}}^{\top}, 1]^{\top}$, the normalized position error $\boldsymbol{s}_{\mathrm{e}}$ is defined as $\boldsymbol{s}_{\mathrm{e}} = \boldsymbol{s} - \boldsymbol{s}_{\mathrm{d}}$. Since $\dot{\boldsymbol{s}}_{\mathrm{d}} = \boldsymbol{0}_{3\times 1}$, the visual error kinematics is derived using (15) as:

$$ \dot{\boldsymbol{s}}_{\mathrm{e}} = \dot{\boldsymbol{s}} = \boldsymbol{h} - {}^{\mathcal{V}}\boldsymbol{\omega}_{\mathrm{v}} \times \boldsymbol{s} - \left[\left(\boldsymbol{h} - {}^{\mathcal{V}}\boldsymbol{\omega}_{\mathrm{v}} \times \boldsymbol{s}\right) \cdot \hat{\boldsymbol{e}}_3\right]\boldsymbol{s}, \quad (16) $$

Using (16), the desired optical flow $\boldsymbol{h}_{\mathrm{d}}$ is derived as:

$$ \boldsymbol{h}_{\mathrm{d}} = \dot{\boldsymbol{s}}_{\mathrm{e}} + {}^{\mathcal{V}}\boldsymbol{\omega}_{\mathrm{v}} \times \boldsymbol{s} + \left[h_{\mathrm{rd}} - \left({}^{\mathcal{V}}\boldsymbol{\omega}_{\mathrm{v}} \times \boldsymbol{s}\right) \cdot \hat{\boldsymbol{e}}_3\right]\boldsymbol{s}, \quad (17) $$

where $h_{\mathrm{rd}}$ is the desired radial optical flow, which is a predefined constant. Now, in designing the outer-loop control law, the first objective is to generate suitable optical flow $\boldsymbol{h}_{\mathrm{d}}$ command such that the position error $\boldsymbol{s}_{\mathrm{e}}$ asymptotically converges to the origin when subjected to the visibility constraint (14).

Using $\boldsymbol{h}_{\mathrm{d}}$ from (17), the optical flow error $\boldsymbol{h}_{\mathrm{e}}$ is estimated as $\boldsymbol{h}_{\mathrm{e}} = \boldsymbol{h} - \boldsymbol{h}_{\mathrm{d}}$. Then, the optical flow error dynamics is given as:

$$ \dot{\boldsymbol{h}}_{\mathrm{e}} = \beta\boldsymbol{u}_{\mathrm{a}} + \boldsymbol{c}_h + \boldsymbol{d}_h, \quad (18) $$

where

$$ \boldsymbol{u}_{\mathrm{a}} = -{}^{\mathcal{I}}R_{\mathcal{V}}^{\top}\,{}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}},\ \beta = \frac{1}{z_{\mathrm{t}}} $$

Note that the function $\boldsymbol{c}_h$ is known and can be estimated for feedback linearization. Whereas, parameter $\beta$ and function $\boldsymbol{d}_h$ are unknown.

*Assumption 1*: Given that $\boldsymbol{u}_\mathrm{a}$ is bounded, the disturbance $\boldsymbol{d}_h$ in (18) is a uniformly bounded function with an unknown upper bound vector $\boldsymbol{d}_\mathrm{max}$, i.e. $|\boldsymbol{d}_h| \leq \boldsymbol{d}_\mathrm{max}$. Moreover, the parameter $\beta$ satisfies the inequality $0 < \beta_\mathrm{min} \leq \beta \leq \beta_\mathrm{max}$, where the constants $\beta_\mathrm{min}$, $\beta_\mathrm{max}$ exist but are unknown.

Now, in designing the outer-loop control law, the second objective is to generate suitable control signal $\boldsymbol{u}_\mathrm{a}$, and hence the desired acceleration ${}^{\mathcal{I}}\boldsymbol{a}_\mathrm{d}$ command, such that the optical flow error $\boldsymbol{h}_\mathrm{e}$ asymptotically converges to the origin in the presence of uncertainty parameter $\beta$ and disturbance term $\boldsymbol{d}_h$. Then, the desired acceleration ${}^{\mathcal{I}}\boldsymbol{a}_\mathrm{d}$ is used to estimate desired roll ($\phi_\mathrm{d}$), pitch ($\theta_\mathrm{d}$), and total thrust (${}^{\mathcal{B}}T_u$) commands using (8).

In order to achieve orientation alignment without explicit knowledge of the target heading $\psi_t$, the proposed flight controller relies on the target image orientation $\alpha$ defined as:

$$\alpha = \frac{1}{2}\tan^{-1}\left(\frac{2\mu_{11}}{\mu_{20}-\mu_{02}}\right), \tag{19}$$

where $\mu_{ij}$ are the centered moments given as $\mu_{ij} = \sum_{k=1}^{N}(\hat{x}_k - \hat{x})^i(\hat{y}_k - \hat{y})^j$. Using (19), the time derivative of $\alpha$ is given as:

$$\dot{\alpha} = -{}^{\mathcal{V}}\omega_{\mathrm{v}z} \tag{20}$$

where ${}^{\mathcal{V}}\omega_{\mathrm{v}z}$ is the third component of the angular rate ${}^{\mathcal{V}}\boldsymbol{\omega}_\mathrm{v}$ of the virtual plane in the virtual frame $\mathcal{V}$. Since ${}^{\mathcal{V}}\boldsymbol{\omega}_\mathrm{v} = [0, 0, -\dot{\psi}]^\top$, this implies that $\dot{\alpha} = \dot{\psi}$.

Hence, the outer-loop of the proposed flight controller passes the desired attitude command given by $\boldsymbol{\Phi}_\mathrm{d} = [\phi_\mathrm{d}, \theta_\mathrm{d}, \alpha_\mathrm{d}]^\top$ to the inner-loop controller to regulates the UAV attitude given by $\boldsymbol{\Phi}_\mathrm{b} = [\phi_\mathrm{b}, \theta_\mathrm{b}, \alpha]^\top$. Note that the inner-loop control architecture of the proposed flight controller mirrors that of the pose-based controller, given by (9)-(12), but replaces UAV heading $\psi$ with target image orientation $\alpha$ to achieve orientation alignment without explicit knowledge of the target heading $\psi_t$.

## III. Formulation of Control Strategy

The proposed flight controller replaces the scale-dependent state variables ${}^{\mathcal{I}}\boldsymbol{r}_\mathrm{b}$ and ${}^{\mathcal{I}}\boldsymbol{v}_\mathrm{b}$, given in (6), with the scale-independent target parameters ${}^{\mathcal{V}}\boldsymbol{s}$ and ${}^{\mathcal{V}}\boldsymbol{h}$. As mentioned in the previous section, the adoption of these target parameters poses challenges, rendering conventional PID controllers unsuitable.

### A. Performance Control for Visibility Constraints

Based on the visual error kinematics (16) developed in the previous section, the prescribed performance control technique is utilized here to achieve predefined dynamic and steady-state response for visual error, as well as satisfy the visibility constraints (14). Defining image error as $\hat{\boldsymbol{r}}_\mathrm{e} = {}^{\mathcal{V}}\hat{\boldsymbol{r}} - \hat{\boldsymbol{r}}_\mathrm{d}$, the visibility constraint (14) can be rewritten in terms of $\hat{\boldsymbol{r}}_\mathrm{e}$ as:

$$-\frac{\mathcal{R}}{2f} - \hat{\boldsymbol{r}}_\mathrm{d} \leq \hat{\boldsymbol{r}}_\mathrm{e} \leq \frac{\mathcal{R}}{2f} - \hat{\boldsymbol{r}}_\mathrm{d}, \tag{21}$$

A controller achieves the prescribed performance when the image error $\hat{\boldsymbol{r}}_\mathrm{e}$ converges to a predefined residual set that is bounded by absolutely decaying function of time. Typically, the prescribed performance can be mathematically expressed by the following inequalities:

$$-\underline{C}_k p_{1k}(t) < \hat{r}_{\mathrm{e}k} < \overline{C}_k p_{1k}(t), \tag{22}$$

where $k = \{1, 2\}$, $\underline{C}_k$ and $\overline{C}_k$ are positive parameters, and the performance function $p_{1k}(t)$ is defined by:

$$p_{1_k}(t) = \left(1 - \frac{p_{1_\infty k}}{\max\{\underline{C}_k, \overline{C}_k\}}\right)e^{-\xi_{1_k}t} + \frac{p_{1_\infty k}}{\max\{\underline{C}_k, \overline{C}_k\}}, \tag{23}$$

where the constant $\xi_{1_k} > 0$ denotes the convergence rate of the tracking error, and $p_{1_\infty k} > 0$ represents the steady-state performance specification, which can be set small enough to meet performance requirements. The visibility constraint (21) is ensured by selecting $\underline{C}_k$ and $\overline{C}_k$ as:

$$\underline{C} = \frac{\mathcal{R}}{2f} + \hat{\boldsymbol{r}}_\mathrm{d},\ \overline{C} = \frac{\mathcal{R}}{2f} - \hat{\boldsymbol{r}}_\mathrm{d}, \tag{24}$$

To achieve target tracking with prescribed performance (22), an error transformation first is introduced. The transformation function $\mathcal{S}_1(\boldsymbol{\zeta}_1)$ is defined such that

$$\hat{\boldsymbol{r}}_\mathrm{e} = \mathcal{S}_1(\boldsymbol{\zeta}_1)\boldsymbol{p}_1(t),\ \mathcal{S}_1(\boldsymbol{\zeta}_1) = \mathrm{diag}(S_1(\zeta_{1_k})), \tag{25}$$

where function $S_1(\zeta_{1_k})$ is strictly increasing and satisfies the following requirements:

$$\begin{aligned}
&-\underline{C}_k < S_1(\zeta_{1_k}) < \overline{C}_k,\\
&\lim_{\zeta_{1_k}\to 0} S_1(\zeta_{1_k}) = 0,\\
&\lim_{\zeta_{1_k}\to \infty} S_1(\zeta_{1_k}) = \overline{C}_k,\\
&\lim_{\zeta_{1_k}\to -\infty} S_1(\zeta_{1_k}) = -\underline{C}_k.
\end{aligned}$$

According to the above mentioned requirements, the error transformation function $S_1(\zeta_{1_k})$ can be chosen as:

$$S_1(\zeta_{1_k}) = \left(\frac{\underline{C}_k\overline{C}_k(e^{\zeta_{1_k}} - 1)}{\underline{C}_k e^{\zeta_{1_k}} + \overline{C}_k}\right). \tag{26}$$

Since $S_1(\zeta_{1_k})$ is strictly increasing, its inverse function always exists. The inverse function $S_1^{-1}(\hat{r}_{\mathrm{e}k}, p_{1_k})$ can be described by:

$$S_1^{-1}(\hat{r}_{\mathrm{e}k}, p_{1_k}) = \zeta_{1_k} = \ln\left(\frac{\overline{C}_k\hat{r}_{\mathrm{e}k} + \underline{C}_k\overline{C}_k p_{1_k}(t)}{\underline{C}_k\overline{C}_k p_{1k}(t) - \underline{C}_k\hat{r}_{\mathrm{e}k}}\right). \tag{27}$$

The time-derivative of transformed error $\zeta_{1_k}$ is given as:

$$\dot{\boldsymbol{\zeta}}_1 = \mathcal{G}_1\left[\dot{\hat{\boldsymbol{r}}}_\mathrm{e} - \mathcal{S}_1(\boldsymbol{\zeta}_1)\dot{\boldsymbol{p}}_1(t)\right], \tag{28}$$

where

$$\mathcal{G}_1 = \left[\boldsymbol{p}_1(t)^\top\frac{\partial\mathcal{S}_1(\boldsymbol{\zeta}_1)}{\partial\boldsymbol{\zeta}_1}\right]^{-1} = \mathrm{diag}(g_{1_k}(\zeta_{1_k}, p_{1_k})),$$

$$g_{1_k}(\zeta_{1_k}, p_{1_k}) = \frac{(\underline{C}_k e^{\zeta_{1_k}} + \overline{C}_k)^2}{\underline{C}_k\overline{C}_k(\overline{C}_k + \underline{C}_k)\,e^{\zeta_{1_k}}p_{1_k}(t)}.$$

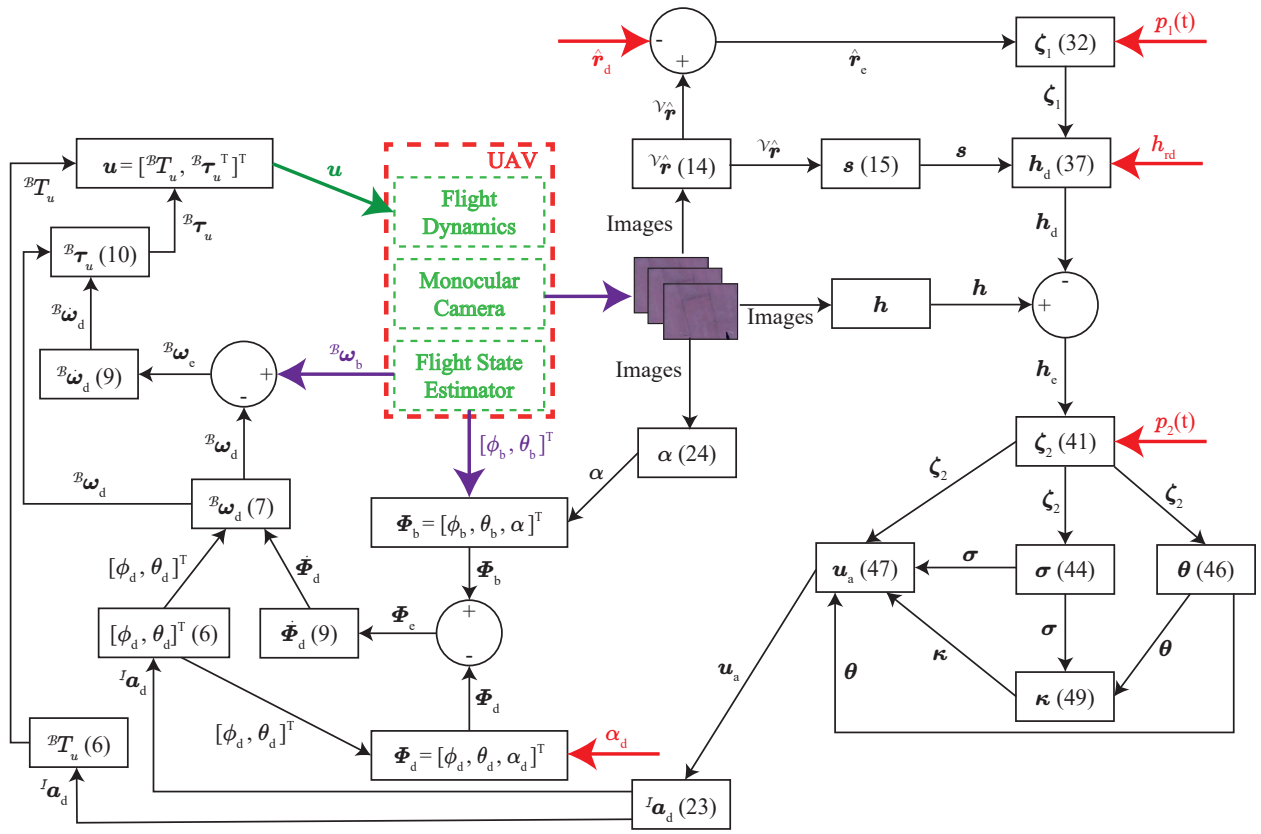


Fig. 1: Block diagram illustrating the control pipeline for soft precise landing of a UAV on a mobile target.

Now, the the visual error kinematics on the unconstrained error is considered. Hence, using (28), $\dot{s}_e$ can be analytically computed as:

$$\dot{s}_e = \begin{bmatrix} \dot{\hat{r}}_e \\ 0 \end{bmatrix} = \begin{bmatrix} \mathcal{G}_1^{-1}\dot{\zeta}_1 + \mathcal{S}_1(\zeta_1)\dot{p}_1(t) \\ 0 \end{bmatrix}. \tag{29}$$

Substituting (29) in (17), we obtain

$$h_d = \begin{bmatrix} \mathcal{G}_1^{-1}\dot{\zeta}_1 + \mathcal{S}_1(\zeta_1)\dot{p}_1(t) \\ 0 \end{bmatrix} + {}^{\mathcal{V}}\omega_v \times s + \left[h_{rd} - \left({}^{\mathcal{V}}\omega_v \times s\right)\cdot \hat{e}_3\right] s, \tag{30}$$

So, the first objective translates to the formulation of a control law that synthesizes suitable optical flow $h_d$ command such that the unconstrained error $\zeta_1$ asymptotically converges to the origin. To this end, a simple PID control law is proposed as given below:

$$\dot{\zeta}_1 = -K_{\zeta p}\zeta_1 - K_{\zeta i}\int \zeta_1 dt - K_{\zeta d}\frac{\delta\zeta_1}{\delta t}. \tag{31}$$

Substituting (31) in (30), we obtain:

$$h_d = \begin{bmatrix} -\mathcal{G}_1^{-1}\left(K_{\zeta p}\zeta_1 + K_{\zeta i}\int \zeta_1 dt + K_{\zeta d}\frac{\delta\zeta_1}{\delta t}\right) + \mathcal{S}_1(\zeta_1)\dot{p}_1(t) \\ 0 \end{bmatrix} + {}^{\mathcal{V}}\omega_v \times s + \left[h_{rd} - \left({}^{\mathcal{V}}\omega_v \times s\right)\cdot \hat{e}_3\right] s. \tag{32}$$

## B. Performance Function for Optical Flow

Based on the perturbed error dynamics (18) developed in the previous section, the prescribed performance control technique is utilized here to achieve predefined dynamic and steady-state response for optical flow error, as well as ensure the boundedness during landing. To this end, the optical flow error $h_e$ is constrained as:

$$h_{e_k} \in (-p_{2_k}(t), p_{2_k}(t)), \tag{33}$$

where $k = \{1, 2, 3\}$ and the performance function $p_2(t))$ is defined by:

$$p_2(t) = e^{-\Xi_i t}(p_{2_0} - p_{2_\infty}) + p_{2_\infty} \tag{34}$$

such that $\lim_{t\to\infty} p_{2_k}(t) = p_{2_\infty k} > 0$. Here, $p_{2_0 k}$, $p_{2_\infty k}$, and matrix $\Xi_2 = \mathrm{diag}(\xi_{2_k})$ are predefined positive constants, which represent the maximum allowable overshoot, the steady-state residual set and the minimum convergence rate of $h_e$, respectively.

A non-linear transformation is formulated to map the constrained output error $h_e$ onto a new unconstrained variable $\zeta_2$ such that:

$$h_e = \mathcal{S}_2(\zeta_2)p_2(t),\ \mathcal{S}_2(\zeta_2) = \mathrm{diag}(S_2(\zeta_{2_k})),$$
$$S_2(\zeta_{2_k}) = \left(\frac{e^{\zeta_{2_k}} - 1}{e^{\zeta_{2_k}} + 1}\right) \tag{35}$$

with $S_2(\zeta_{2_k})$ possessing the following properties:

I. $-1 < S_2(\zeta_{2_k}) < 1,\ \forall\ \zeta_{2_k}(t) \in \mathbb{R}$,
II. $\lim_{\zeta_{2_k}\to\infty} S_2(\zeta_{2_k}) = 1$,
III. $\lim_{\zeta_{2_k}\to-\infty} S_2(\zeta_{2_k}) = -1$

Moreover, the components of the unconstrained output $\boldsymbol{\zeta}_2$ can be expressed as:

$$\zeta_{2_k} = S_2^{-1}(h_{e_k}, p_{2_k}) = ln\left(\frac{p_{2_k} + h_{e_k}}{p_{2_k} - h_{e_k}}\right). \quad (36)$$

Note that the inverse transformation $\mathcal{S}_2^{-1} : (-1, 1) \to \mathbb{R}$ is well-defined, provided $\zeta_{2_k}(t) \in \mathbb{R}$.

Differentiating Eq. (35) and rearranging, we have:

$$\dot{\boldsymbol{\zeta}}_2 = \mathcal{G}_2\big[\dot{\boldsymbol{h}}_e - \mathcal{S}_2(\boldsymbol{\zeta}_2)\dot{\boldsymbol{p}}_2(t)\big], \quad (37)$$

where

$$\mathcal{G}_2 = \left[\boldsymbol{p}_2(t)^\top \frac{\partial \mathcal{S}_2(\boldsymbol{\zeta}_2)}{\partial \boldsymbol{\zeta}_2}\right]^{-1} = \text{diag}(g_{2_k}(\zeta_{2_k}, p_{2_k}),$$

$$g_{2_k}(\zeta_{2_k}, p_{2_k}) = \frac{(e^{\zeta_{2_k}} + 1)^2}{2\, e^{\zeta_{2_k}} p_{2_k}(t)}$$

Substituting (18) in the above equation (37) to derive an unconstrained perturbed system equation enveloped with prescribed performance as shown below:

$$\dot{\boldsymbol{\zeta}}_2 = \mathcal{G}_2\big[\beta \boldsymbol{u}_\text{a} + \boldsymbol{c}_h + \boldsymbol{d}_h - \mathcal{S}_2(\boldsymbol{\zeta}_2)\dot{\boldsymbol{p}}_2(t)\big] \quad (38)$$

## C. Adaptive Control Law

The second objective translates to the formulation of a control law that synthesizes suitable control signal $\boldsymbol{u}_\text{a}(t)$ to drive the non-linear map-based transformed output $\boldsymbol{\zeta}_2$ to a small uniform ultimate bound (UUB) around the origin in finite time through online estimation and rejection of unknown bounded uncertainty parameters $\beta$, $\boldsymbol{d}_h$ shown in Eq. (38). To this end, an adaptive first-order sliding mode control (AFOSMC) strategy is designed that compensates for the system uncertainty without any a priori knowledge of its upper bound. This control strategy accounts for non-linearity and singularity inherent to optical flow-guided landing and alleviates the adverse effects of actuator faults and input saturation.

Similar to the outer loop controller, the following first-order PI sliding surface $\boldsymbol{\sigma}$ is defined for the inner loop output error vector $\boldsymbol{\zeta}_2$:

$$\boldsymbol{\sigma} = \boldsymbol{\zeta}_2 + \mathcal{X}\int_{t_0}^{t} \boldsymbol{\zeta}_2 \; d\tau, \quad (39)$$

where the sliding surface $\sigma_k$ represents the exponentially converging dynamics of output error $\zeta_{2_k}(t)$ with a decaying rate $\chi_{2_k} > 0$ for $k = \{1, 2, 3\}$ such that $\mathcal{X} = \text{diag}(\chi_k)$. The time-derivative of $\boldsymbol{\sigma}$ can be computed as:

$$\dot{\boldsymbol{\sigma}} = \dot{\boldsymbol{\zeta}}_2 + \mathcal{X}\;\boldsymbol{\zeta}_2 \quad (40)$$

The following equation can be obtained by substituting (38) in the above Eq. (40):

$$\begin{aligned}\dot{\boldsymbol{\sigma}} &= \mathcal{G}_2\big[\beta \boldsymbol{u}_\text{a} + \boldsymbol{c}_h + \boldsymbol{d}_h - \mathcal{S}_2(\boldsymbol{\zeta}_2)\dot{\boldsymbol{p}}_2(t)\big] + \mathcal{X}\;\boldsymbol{\zeta}_2 \\ &= \beta\mathcal{G}_2\big[\boldsymbol{u}_\text{a} + \boldsymbol{c}_h - \mathcal{S}_2(\boldsymbol{\zeta}_2)\dot{\boldsymbol{p}}_2(t) + \mathcal{G}_2^{-1}\mathcal{X}\,\boldsymbol{\zeta}_2\big] + \beta_\text{min}\,\mathcal{G}_2\boldsymbol{\theta}\overline{\boldsymbol{d}},\end{aligned} \quad (41)$$

where

$$\boldsymbol{\theta} = \big[-\boldsymbol{c}_h + \mathcal{S}_2(\boldsymbol{\zeta}_2)\dot{\boldsymbol{p}}_2(t) - \mathcal{G}_2^{-1}\mathcal{X}\,\boldsymbol{\zeta}_2 \quad I_{3\times3}\big],$$

$$\overline{\boldsymbol{d}} = \beta_\text{min}^{-1}\;\big[\beta - 1 \quad \boldsymbol{d}_h\big]^\top,$$

There exists an unknown positive constant $\tilde{d} = \beta_\text{min}^{-1}\Big[\max\{|\beta_\text{max} - 1|, |\beta_\text{min} - 1|\} + \|\boldsymbol{d}_\text{max}\|\Big]$ such that $\|\overline{\boldsymbol{d}}\| \le \tilde{d}$.

Finally, the following control law has been proposed:

$$\begin{aligned}\boldsymbol{u}_\text{a} &= \mathcal{G}_2^{-1}\big[\boldsymbol{u}_\text{sw} + \boldsymbol{u}_\text{eq}\big], \\ \boldsymbol{u}_\text{sw} &= -\Gamma\boldsymbol{\sigma} - \|\boldsymbol{\theta}\|\,\text{sat}(\mathcal{E}^{-1}\boldsymbol{\sigma})\mathcal{G}_2\boldsymbol{\kappa}(t) \\ \boldsymbol{u}_\text{eq} &= \mathcal{G}_2\big[-\boldsymbol{c}_h + \mathcal{S}_2(\boldsymbol{\zeta}_2)\dot{\boldsymbol{p}}_2(t) - \mathcal{G}_2^{-1}\mathcal{X}\,\boldsymbol{\zeta}_2\big],\end{aligned} \quad (42)$$

where $\Gamma = \text{diag}(\gamma_k)$ is the proportional control matrix with positive constants $\gamma_k$ and $\boldsymbol{\kappa}(t)$ is the adaptive sliding mode control parameter. The design matrix $\mathcal{E}_i$ is defined as $\mathcal{E} = \text{diag}(\varepsilon_k)$ such that $0 < \varepsilon_k << 1$, where the design constants $\varepsilon_k$ represent the sensitivity of the smooth transition across the sliding surface for sliding mode control. the sliding surface for sliding mode control. The saturation matrix $\text{sat}(\mathcal{E}^{-1}\boldsymbol{\sigma}) = \text{diag}(\text{sat}(\sigma_k/\varepsilon_k))$, where the saturation function $\text{sat}(\sigma_k/\epsilon_k)$ is defined as

$$\text{sat}(\sigma_k/\epsilon_k) = \begin{cases}\sigma_k/\epsilon_k, & \text{if } |\sigma_k| < \epsilon_k \\ \lceil\sigma_k\rfloor^0, & \text{otherwise}\end{cases} \quad (43)$$

with $\lceil\sigma_k\rfloor^0$ representing the sign function. The adaptive gain $\boldsymbol{\kappa}(t)$ is updated as:

$$\dot{\boldsymbol{\kappa}}(t) = \|\boldsymbol{\theta}\|\mathcal{N}\mathcal{G}_2|\boldsymbol{\sigma}| - \mathcal{N}\mathcal{P}\boldsymbol{\kappa}(t),\; \boldsymbol{\kappa}(0) \in \mathbb{R}_+^3, \quad (44)$$

where $|\boldsymbol{\sigma}| = [|\sigma_1|, |\sigma_2|, |\sigma_3|]^\top$. Also, $\mathcal{N} = \text{diag}(\eta_k)$ and $\mathcal{P} = \text{diag}(\rho_k)$ are positive gain matrices.

Eq. (42) is incorporated in Eq. (41) to obtain the following expression:

$$\dot{\boldsymbol{\sigma}} = -\beta\big[\Gamma\boldsymbol{\sigma} + \|\boldsymbol{\theta}\|\,\text{sat}(\mathcal{E}^{-1}\boldsymbol{\sigma})\mathcal{G}_2\boldsymbol{\kappa}(t)\big] + \beta_\text{min}\,\mathcal{G}_2\boldsymbol{\theta}\overline{\boldsymbol{d}}, \quad (45)$$

## D. Lyapunov Stability Analysis

The stability analysis starts with the following definition of global uniform ultimate boundedness of the signal $\boldsymbol{\chi}(t)$:

*Definition 1*: Consider a multi-variable signal $\boldsymbol{\chi}(t) \in \mathbb{R}^n,\; n \in [0, \infty)$. Assume that $\exists\; \lambda,\; \lambda_\text{u} \in \mathbb{R}_+$, where $\lambda,\; \lambda_\text{u}$ are positive constants and $\forall\, 0 < \lambda < \lambda_\text{u}$, $\exists\; \mathcal{T} \ge 0$ and $\Lambda > 0$ such that

$$\|\boldsymbol{\chi}(0)\| \le \lambda \implies \|\boldsymbol{\chi}(t)\| \le \Lambda,\, \forall\, t \ge \mathcal{T} \quad (46)$$

and $\lambda$ can be arbitrarily large. Then, the signal $\boldsymbol{\chi}(t)$ is said to be *global uniformly ultimately bounded* by the ultimate bound $\Lambda$.

*Lemma 1*: Let $V$ be a continuously differentiable function such that

$$k_1\|\boldsymbol{\chi}\|_2^2 \le V(\boldsymbol{\chi}) \le k_2\|\boldsymbol{\chi}\|_2^2,\; \dot{V}(\boldsymbol{\chi}) \le -k_3 V(\boldsymbol{\chi}), \quad \forall\, \|\boldsymbol{\chi}\| \ge \varkappa \quad (47)$$

with $k_1,\; k_2,\; k_3 > 0$. Consider a ball $\mathcal{B}_r$ of radius $r$ and let $\varkappa \le r(k_1/k_2)^{\frac{1}{2}} \triangleq \lambda$. Then, for every initial state $\boldsymbol{\chi}(0)$ satisfying $\|\boldsymbol{\chi}(0)\|_2^2 \le \lambda$, $\exists\; \mathcal{T} \ge 0$, such that (46) holds true with $\Lambda \triangleq \varkappa(k_2/k_1)^{\frac{1}{2}}$ and an arbitrarily large $\lambda$.

*Theorem 1*: Consider the perturbed nonlinear system (18) such that the initial value of the optical flow error

$\boldsymbol{h}_e(0)$ is within the constrained region, i.e., $-\boldsymbol{p}_{2_0} < \boldsymbol{h}_e(0) < \boldsymbol{p}_{2_0}$. Then, under *Assumption 1*, system (38) under the proposed control law (42) accomplishes exponential convergence of $\boldsymbol{\zeta}_2$ to a UUB given by

$$\vartheta = \sqrt{\frac{3\beta_{\min}\lambda_{\max}(\mathcal{P})}{(\varphi_1 - \varphi_2)}\tilde{d}}, \tag{48}$$

where $\varphi_1 \triangleq \beta_{\min}\frac{\min\left(2\lambda_{\min}(\Gamma),\lambda_{\min}(\mathcal{P})\right)}{\max\left(1,\beta_{\min}\lambda_{\max}(\mathcal{N}^{-1})\right)} > 0$, $0 < \varphi_2 < \varphi_1$. Moreover, the control law (42) ensures that the trajectory of the system (18) lies within the prescribed constraint sets $-\boldsymbol{p}_2(t) < \boldsymbol{h}_e < \boldsymbol{p}_2(t)\,\forall\, t \geq 0$.

*Proof:* As a first step, observe that $\zeta_{2_k}(0) = S_{2_k}^{-1}(h_{e_k}(0)/p_{2_0k})$ is well-defined for $|h_{e_k}(0)| < p_{2_0k}$ for $k = \{1, 2, 3\}$. The dynamics of $\boldsymbol{\zeta}_2$ then evolves according to the system (38) under the control law (42) of the PLASMC strategy.

The following candidate Lyapunov function $V(t)$ is now considered for subsequent analysis:

$$V(t) = \frac{1}{2}\boldsymbol{\sigma}^\top\boldsymbol{\sigma} + \frac{\beta_{\min}}{2}\left[\boldsymbol{\kappa}(t) - \tilde{d}\mathbf{1}\right]^\top \mathcal{N}^{-1}\left[\boldsymbol{\kappa}(t) - \tilde{d}\mathbf{1}\right], \tag{49}$$

where the unknown positive constant $\tilde{d}$ satisfies $\|\overline{\boldsymbol{d}}\| \leq \tilde{d}$ and $\mathbf{1} = [1,\, 1,\, 1]^\top$.

Using (45) and (44), the time-derivative of (49) can be written as

$$\begin{aligned}\dot{V}(t) &= \boldsymbol{\sigma}^\top\dot{\boldsymbol{\sigma}} + \beta_{\min}\left[\boldsymbol{\kappa}(t) - \tilde{d}\mathbf{1}\right]^\top \mathcal{N}^{-1}\dot{\boldsymbol{\kappa}}(t)\\ &= -\beta\,\boldsymbol{\sigma}^\top\left[\Gamma\boldsymbol{\sigma} + \|\boldsymbol{\theta}\|\,\mathrm{sat}(\mathcal{E}^{-1}\boldsymbol{\sigma})\mathcal{G}_2\boldsymbol{\kappa}(t)\right]\\ &\quad + \beta_{\min}\,\boldsymbol{\sigma}^\top\mathcal{G}_2\boldsymbol{\theta}\overline{\boldsymbol{d}}\\ &\quad + \beta_{\min}\left[\boldsymbol{\kappa}(t) - \tilde{d}\mathbf{1}\right]^\top \mathcal{N}^{-1}\left[\|\boldsymbol{\theta}\|\mathcal{N}\mathcal{G}_2|\boldsymbol{\sigma}| - \mathcal{N}\mathcal{P}\boldsymbol{\kappa}(t)\right]\end{aligned} \tag{50}$$

Since the diagonal matrices $\Gamma$ is positive-definite, it can be shown that $\boldsymbol{\sigma}^\top\Gamma\boldsymbol{\sigma} > 0$. Moreover, $\boldsymbol{\sigma}^\top\mathrm{sat}(\mathcal{E}^{-1}\boldsymbol{\sigma}) \approx |\boldsymbol{\sigma}|^\top \in \mathbb{R}_+^3$ and $\boldsymbol{\kappa}(t) \in \mathbb{R}_+^3$ from Eq. (44), which implies $|\boldsymbol{\sigma}|^\top\mathcal{G}_2\boldsymbol{\kappa}(t) > 0$. Since $\beta \geq \beta_{\min} > 0$, (50) can be rewritten as:

$$\begin{aligned}\dot{V}(t) \leq\ & -\beta_{\min}\boldsymbol{\sigma}^\top\Gamma\boldsymbol{\sigma} - \beta_{\min}\|\boldsymbol{\theta}\||\boldsymbol{\sigma}|^\top\mathcal{G}_2\boldsymbol{\kappa}(t)\\ & + \beta_{\min}\|\overline{\boldsymbol{d}}\|\|\boldsymbol{\theta}\||\boldsymbol{\sigma}|^\top\mathcal{G}_2\mathbf{1}\\ & + \beta_{\min}\left[\boldsymbol{\kappa}(t) - \tilde{d}\mathbf{1}\right]^\top\left[\|\boldsymbol{\theta}\|\mathcal{G}_2|\boldsymbol{\sigma}| - \mathcal{P}\boldsymbol{\kappa}(t)\right]\\ \leq\ & -\beta_{\min}\boldsymbol{\sigma}^\top\Gamma\boldsymbol{\sigma} - \beta_{\min}\left(\tilde{d} - \|\overline{\boldsymbol{d}}\|\right)\|\boldsymbol{\theta}\||\boldsymbol{\sigma}|^\top\mathcal{G}_2\mathbf{1}\\ & - \beta_{\min}\left[\boldsymbol{\kappa}^\top(t)\mathcal{P}\boldsymbol{\kappa}(t) - \tilde{d}\mathbf{1}^\top\mathcal{P}\boldsymbol{\kappa}(t)\right]\end{aligned} \tag{51}$$

Given that $\|\overline{\boldsymbol{d}}\| \leq \tilde{d}$, it can be shown that $\beta_{\min}\left(\tilde{d} - \|\overline{\boldsymbol{d}}\|\right)\|\boldsymbol{\theta}\||\boldsymbol{\sigma}|^\top\mathcal{G}_2\mathbf{1} \geq 0$. Moreover, since $\mathcal{P}$ is a positive-definite matrix, $\beta_{\min}\left[\boldsymbol{\kappa}^\top(t)\mathcal{P}\boldsymbol{\kappa}(t) - \tilde{d}\mathbf{1}^\top\mathcal{P}\boldsymbol{\kappa}(t)\right] > \frac{\beta_{\min}}{2}\left\{\left[\boldsymbol{\kappa}(t) - \tilde{d}\mathbf{1}\right]^\top\mathcal{P}\left[\boldsymbol{\kappa}(t) - \tilde{d}\mathbf{1}\right] - \tilde{d}^2\mathbf{1}^\top\mathcal{P}\mathbf{1}\right\}$. Hence,

$$\begin{aligned}\dot{V}(t) \leq\ & -\beta_{\min}\lambda_{\min}(\Gamma)\|\boldsymbol{\sigma}\|^2\\ & - \frac{\beta_{\min}}{2}\left[\boldsymbol{\kappa}(t) - \tilde{d}\mathbf{1}\right]^\top\mathcal{P}\left[\boldsymbol{\kappa}(t) - \tilde{d}\mathbf{1}\right]\\ & + \frac{\beta_{\min}}{2}\tilde{d}^2\mathbf{1}^\top\mathcal{P}\mathbf{1}\\ \leq\ & -\beta_{\min}\Big[\lambda_{\min}(\Gamma)\|\boldsymbol{\sigma}\|^2 + \frac{1}{2}\lambda_{\min}(\mathcal{P})\|\boldsymbol{\kappa}(t) - \tilde{d}\mathbf{1}\|^2\\ & - \frac{3}{2}\lambda_{\min}(\mathcal{P})\tilde{d}^2\Big]\\ \leq\ & -\varphi_1 V(t) + \frac{3}{2}\beta_{\min}\lambda_{\max}(\mathcal{P})\tilde{d}^2,\end{aligned} \tag{52}$$

where $\varphi_1 \triangleq \beta_{\min}\frac{\min\left(2\lambda_{\min}(\Gamma),\lambda_{\min}(\mathcal{P})\right)}{\max\left(1,\beta_{\min}\lambda_{\max}(\mathcal{N}^{-1})\right)} > 0$. For the choice of a scalar $\varphi_2$ satisfying $0 < \varphi_2 < \varphi_1$, the above inequality (52) is rewritten as:

$$\begin{aligned}\dot{V}(t) \leq\ & -\varphi_2 V(t) - (\varphi_1 - \varphi_2)V(t) + \frac{3}{2}\beta_{\min}\lambda_{\max}(\mathcal{P})\tilde{d}^2\\ \leq\ & -\varphi_2 V(t),\ \forall\, V(t) \geq \Theta = \frac{3\beta_{\min}\lambda_{\max}(\mathcal{P})\tilde{d}^2}{2(\varphi_1 - \varphi_2)}\end{aligned} \tag{53}$$

Invoking *Lemma 1*, any trajectory enters the ball $\mathcal{B}_r = \left\{(\boldsymbol{\sigma},\, \boldsymbol{\kappa}) \in \mathbb{R}^8 | V \leq \Theta\right\}$ exponentially in finite-time and stays in $\mathcal{B}_r$ for all future time. Note that the size of the ball can be made suitably small through a suitable selection of the control parameters $\lambda_{\min}(\mathcal{P})$, $\lambda_{\max}(\mathcal{P})$, $\lambda_{\max}(\mathcal{N}^{-1})$ and $\lambda_{\min}(\Gamma)$, so that by recognizing that $V(t) \geq \|\boldsymbol{\sigma}\|^2/2$, one has the ultimate bound for $\boldsymbol{\sigma}$ as $\|\boldsymbol{\sigma}\| \leq \vartheta = [3\beta_{\min}\lambda_{\max}(\mathcal{P})/(\varphi_1 - \varphi_2)]^{\frac{1}{2}}\,\tilde{d}$. This bound is uniform as it is independent of initial conditions. Consequently, it follows from (39) that the output error variable $\boldsymbol{\zeta}_2$ converges to a small bound around the origin so that from (35), $-\boldsymbol{p}_2(t) < \boldsymbol{h}_e < \boldsymbol{p}_2(t)$ is satisfied. Thus the prescribed performance constraint is ensured $\forall\, t \geq 0$. This ends the proof.

| **Run** | ${}^{\mathcal{I}}x$ | ${}^{\mathcal{I}}y$ | ${}^{\mathcal{I}}z$ | Remarks |
|---|---|---|---|---|
| 1 | 0 | 0 | 5 | Nominal |
| 2 | 0 | 0 | 7 | Increase in ${}^{\mathcal{I}}z$ |
| 3 | 0 | 0 | 3 | Decrease in ${}^{\mathcal{I}}z$ |
| 4 | 2 | 2 | 5 | Increase in ${}^{\mathcal{I}}x$, ${}^{\mathcal{I}}y$ |
| 5 | -2 | -2 | 5 | Decrease in ${}^{\mathcal{I}}x$, ${}^{\mathcal{I}}y$ |

TABLE II: Different Initial Conditions for UAV position landing on a target.

## IV. Numerical Simulation Results

This section provides numerical simulation results for soft precise landing on both stationary and mobile platforms. These simulation tests are conducted for different initial conditions as tabulated in *Table* II using the control parameters given in *Table* III.

Moreover, the simulation tests are conducted for five cases, each with different target trajectory as given below:

| Control Parameter | Control Values |
|---|---|
| **Performance Function for Visibility Constraint** | |
| $\Xi_1$ | $\text{diag}([0.5, 0.5])$ |
| $\boldsymbol{p}_{1_\infty}$ | $[0.2, 0.2]^\top$ |
| **PID-based Normalized Position Control** | |
| $K_{\zeta p}$ | $\text{diag}([5.0, 5.0])$ |
| $K_{\zeta i}$ | $\text{diag}([0.5, 0.5])$ |
| $K_{\zeta d}$ | $\text{diag}([0.1, 0.1])$ |
| **Performance Function for Optical Flow** | |
| $\Xi_2$ | $\text{diag}([0.5, 0.5, 0.5])$ |
| $\boldsymbol{p}_{2_0}$ | $[10.0, 10.0, 10.0]^\top$ |
| $\boldsymbol{p}_{2_\infty}$ | $[2.0, 2.0, 2.0]^\top$ |
| **ASMC-based Optical Flow Control** | |
| $\Omega$ | $\text{diag}([0.01, 0.01, 0.01])$ |
| $\Gamma$ | $\text{diag}([0.01, 0.01, 0.01])$ |
| $P$ | $\text{diag}([0.1, 0.1, 0.1])$ |
| $N$ | $\text{diag}([0.01, 0.01, 0.01])$ |
| $\boldsymbol{\kappa}$ | $\text{diag}([0.1, 0.1, 0.1])$ |
| $E$ | $\text{diag}([2.0, 2.0, 2.0])$ |
| **PID-based Attitude Control** | |
| $K_{\Phi p}$ | $\text{diag}([5.0, 5.0, 1.0])$ |
| $K_{\Phi i}$ | $\text{diag}([0.1, 0.1, 0.01])$ |
| $K_{\Phi d}$ | $\text{diag}([0.01, 0.01, 0.01])$ |
| **PID-based Attitude Rate Control** | |
| $K_{\omega p}$ | $\text{diag}([5.0, 5.0, 5.0])$ |
| $K_{\omega i}$ | $\text{diag}([0.1, 0.1, 0.1])$ |
| $K_{\omega d}$ | $\text{diag}([0.04, 0.04, 0.01])$ |

TABLE III: List of all control parameters, along with their values, used in the proposed flight controller

1) *Case 1:* When the target is stationary at position ${}^{\mathcal{I}}\boldsymbol{r}_{\text{t}} = [0.0, 0.0, 0.0]^\top$ m in the inertial frame $\mathcal{I}$.
2) *Case 2:* When the target is moving with a constant linear velocity, ${}^{\mathcal{I}}\boldsymbol{v}_{\text{t}} = [0.3, 0.2, 0.0]^\top$ m/s.
3) *Case 3:* When the target is moving with a non-constant linear velocity in x-direction, ${}^{\mathcal{I}}\boldsymbol{v}_{\text{t}} = [0.5\, sin(0.8\, t), 0.0, 0.0]^\top$ m/s.
4) *Case 4:* When the target is moving with a non-constant linear velocity in x- and y-directions, ${}^{\mathcal{I}}\boldsymbol{v}_{\text{t}} = [0.5\, sin(-0.3\, t), 0.8\, sin(0.5\, t), 0.0]^\top$ m/s.
5) *Case 5:* When the target is undergoing circular motion on a horizontal plane with a turning radius of 1.0 m and turning speed of 0.1 rad/s.

### A. Simulation without Uncertainty and Perturbation

The performance of the proposed flight controller is first assessed under idealized conditions, with varying initial states and no model uncertainties or external disturbances.

### B. Simulation with Uncertainty and Perturbation

In order to simulate more realistic scenarios, the following real-world settings are added in the numerical simulation is conducted in MATLAB under following scenarios:

1) Measurement Noise: The estimated optical flow $\boldsymbol{h}$ is corrupted using additive Gaussian white noise with $\mu$ = 0.1 rad/s and $\sigma$ = 0.05 rad/s.
2) Ground Effect: The ground effect factor is added to the the total rotor control thrust ${}^{\mathcal{B}}T_u$ as given below:
$$ {}^{\mathcal{B}}T_\infty = \frac{1}{1 - \left(\frac{R}{4\,{}^{\mathcal{I}}z_{\text{t}}}\right)^2}\, {}^{\mathcal{B}}T_u, \forall\ t \geq 0, \tag{54} $$
where $R$ is the propeller radius.
3) Computational Delay: A delay of 1 time step is added during the computation of the combined thrust-torque command $\boldsymbol{u} = [\,{}^{\mathcal{B}}T_u,\, {}^{\mathcal{B}}\boldsymbol{\tau}_u^\top]^\top$.
4) Visibility Constraint: The target image error $\hat{\boldsymbol{r}}_{\text{e}}$ is constrained using the performance function $\boldsymbol{p}_1(t)$ as shown in (22).
5) Input Constraint: Given the motor-mixing matrix $M$, the motor thrust vector $\boldsymbol{T}(t)$ is computed as:
$$ \boldsymbol{T}(t) = M^{-1}\boldsymbol{u}, \tag{55} $$
where $T_{\min} \leq T_k(t) \leq T_{\max}$, for motor $k = \{1, 2, 3, 4\}$, with $T_{\min} = 0.01Nm$ and $T_{\max} = 7.0Nm$.

## V. CONCLUSION

This study proposes the development of a novel scale-independent image-based landing controller designed to achieve precise soft landing on both stationary and mobile target surfaces using a robust PLASMC strategy. The key to achieving the functional requirements for soft landing using optic flow is to ensure the convergence and boundedness of the optic flow error to a small bound around the origin. Numerical simulations are conducted to demonstrate the efficacy of the proposed controller in idealistic conditions and its robustness in a real-world setting. These simulation tests are conducted for multiple initial conditions and multiple target trajectories.

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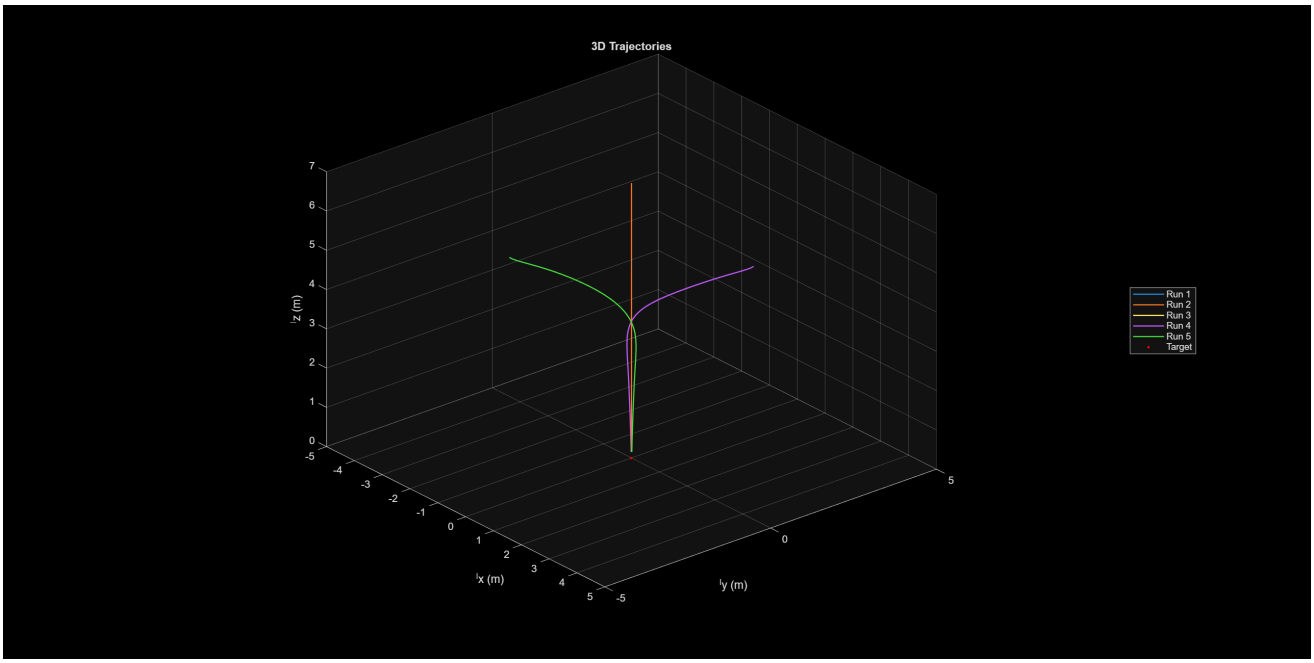


Fig. 2: This 3-D plot shows the evolution of UAV position in inertial frame ${}^{\mathcal{I}}\boldsymbol{r} = [{}^{\mathcal{I}}x, {}^{\mathcal{I}}y, {}^{\mathcal{I}}z]^{\top}$ for different initial conditions (*Table* II) in idealistic conditions, when the target is static on a horizontal plane.

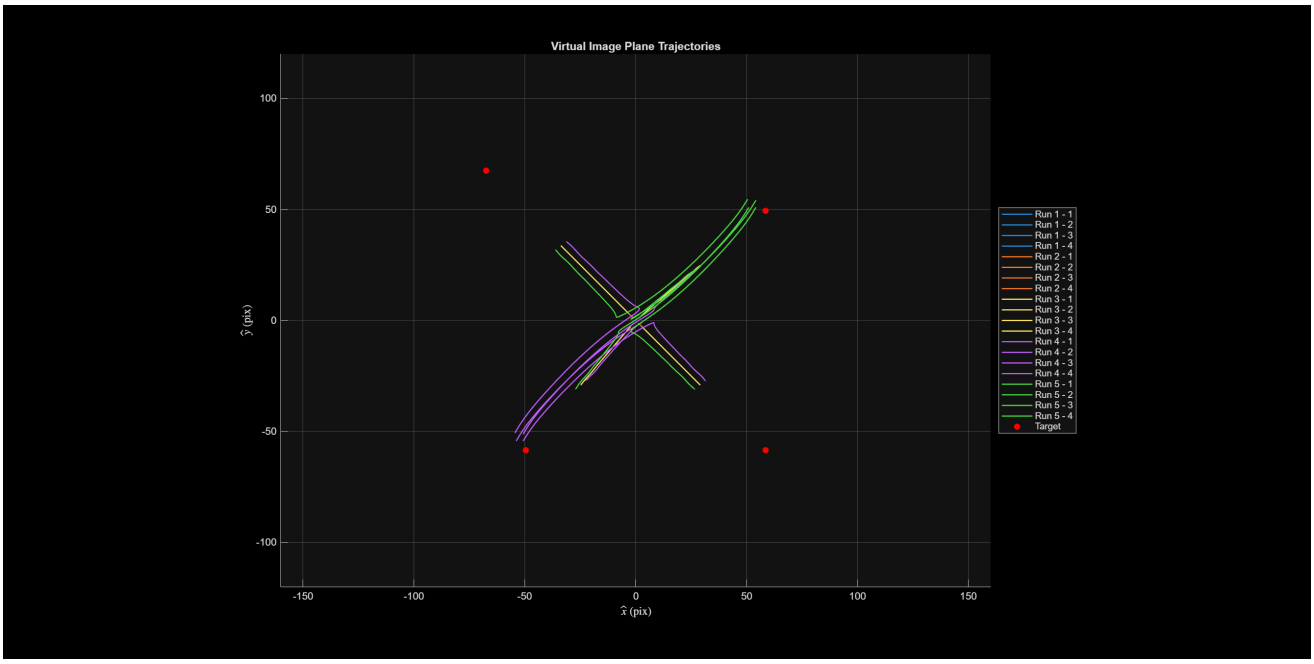


Fig. 3: This graph shows the evolution of virtual image points $\hat{\boldsymbol{r}} = [\hat{x}, \hat{y}]^{\top}$ in the virtual image plane for different initial conditions (*Table* II) in idealistic conditions, when the target is static on a horizontal plane.

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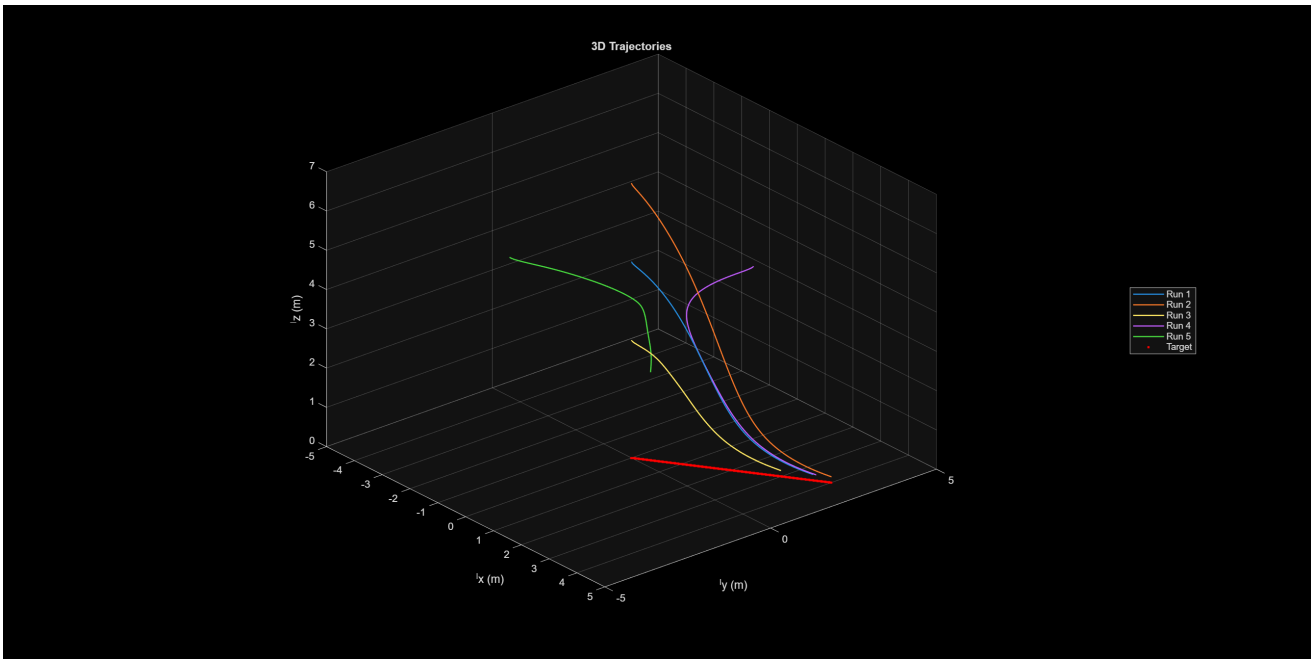


Fig. 4: This 3-D plot shows the evolution of UAV position in inertial frame ${}^{\mathcal{I}}\boldsymbol{r} = [{}^{\mathcal{I}}x, {}^{\mathcal{I}}y, {}^{\mathcal{I}}z]^\top$ for different initial conditions (*Table* II) in idealistic conditions, when the target is undergoing linear motion on a horizontal plane.

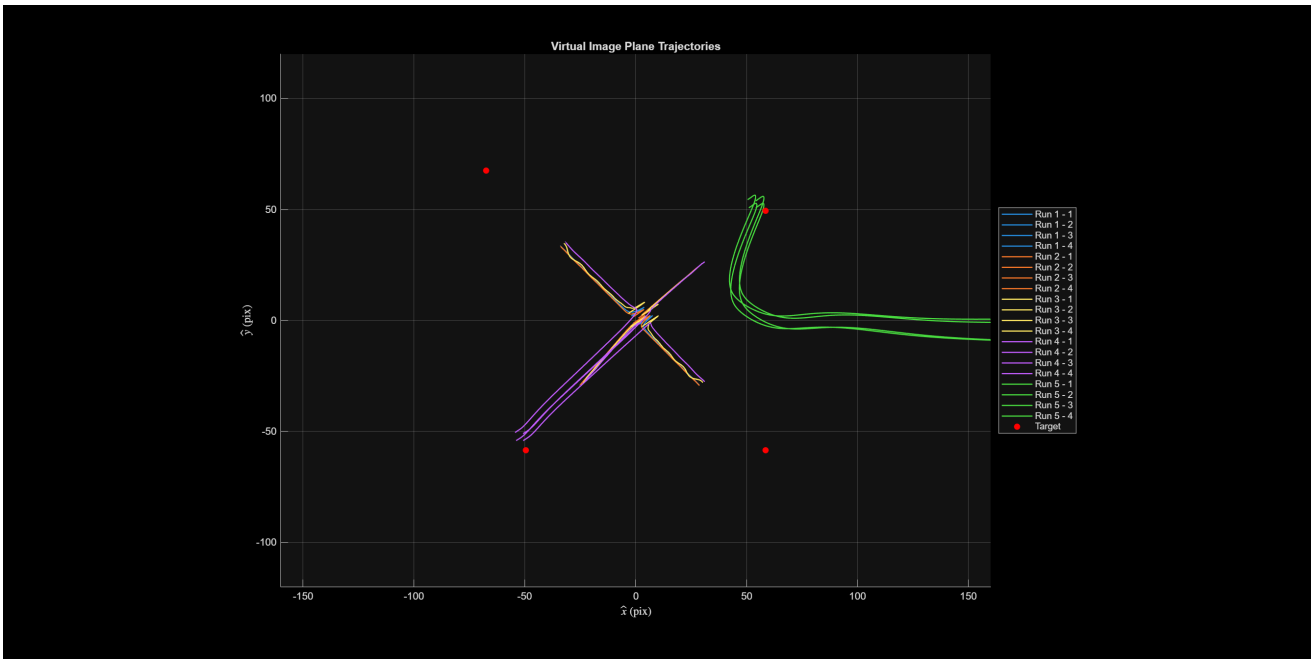


Fig. 5: This graph shows the evolution of virtual image points $\hat{\boldsymbol{r}} = [\hat{x}, \hat{y}]^\top$ in the virtual image plane for different initial conditions (*Table* II) in idealistic conditions, when the target is undergoing linear motion on a horizontal plane.


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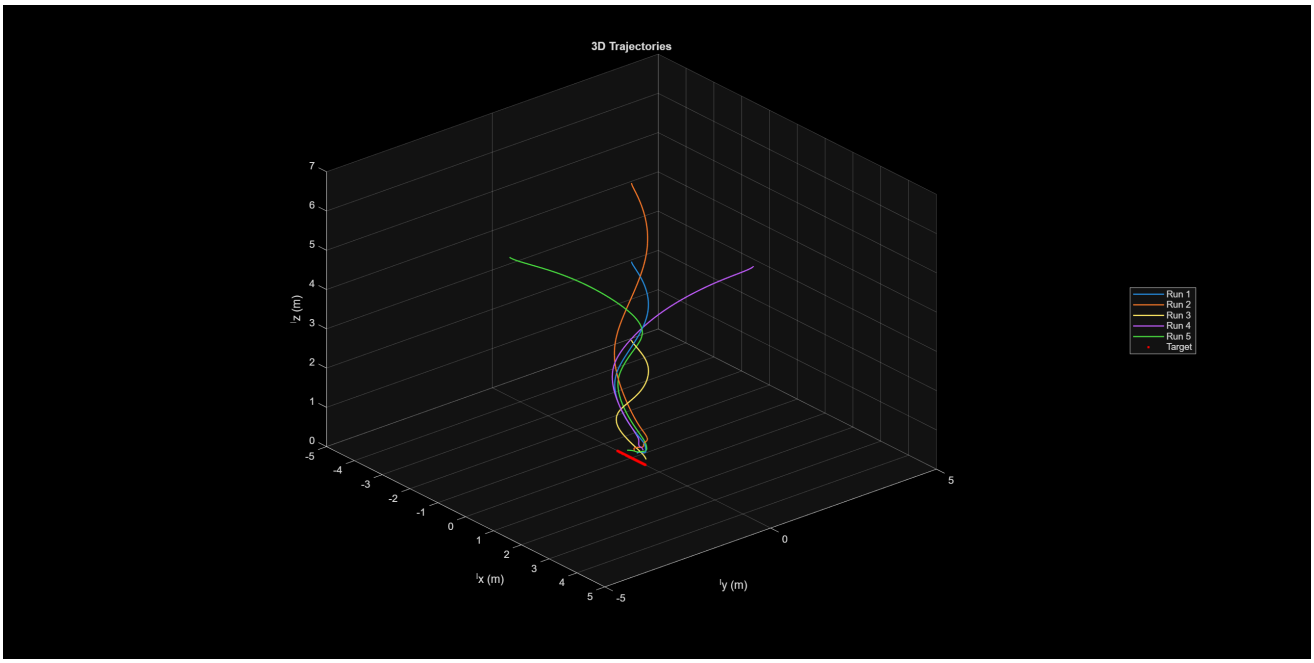


Fig. 6: This 3-D plot shows the evolution of UAV position in inertial frame ${}^{\mathcal{I}}\boldsymbol{r} = [{}^{\mathcal{I}}x, {}^{\mathcal{I}}y, {}^{\mathcal{I}}z]^{\top}$ for different initial conditions (*Table* II) in idealistic conditions, when the target is undergoing sinusoidal motion on a horizontal plane.

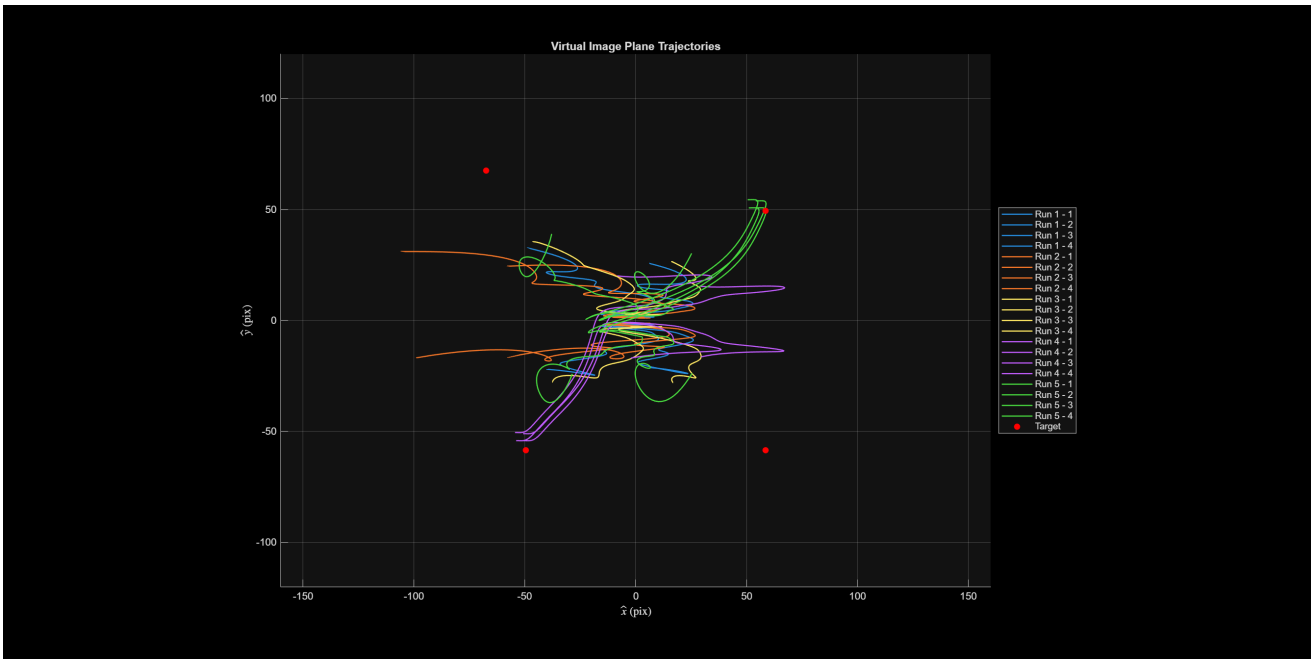


Fig. 7: This graph shows the evolution of virtual image points $\hat{\boldsymbol{r}} = [\hat{x}, \hat{y}]^{\top}$ in the virtual image plane for different initial conditions (*Table* II) in idealistic conditions, when the target is undergoing sinusoidal motion on a horizontal plane.

Adaptive extended disturbance observer-based control for quadrotor landing on fast platform
*IEEE Transactions on Industrial Electronics*, vol. 73, no. 4, pp. 6178–6188, 2026.

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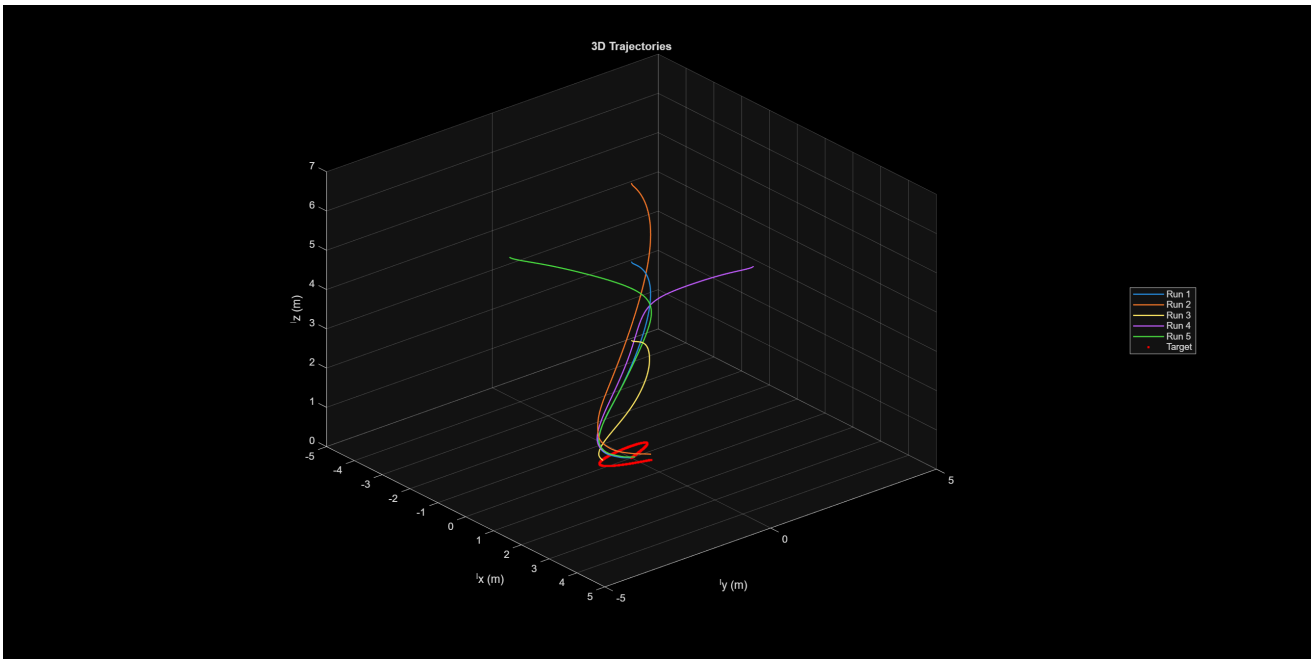


Fig. 8: This 3-D plot shows the evolution of UAV position in inertial frame ${}^{\mathcal{I}}\boldsymbol{r} = [{}^{\mathcal{I}}x, {}^{\mathcal{I}}y, {}^{\mathcal{I}}z]^{\top}$ for different initial conditions (*Table* II) in idealistic conditions, when the target is undergoing lissajous motion on a horizontal plane.

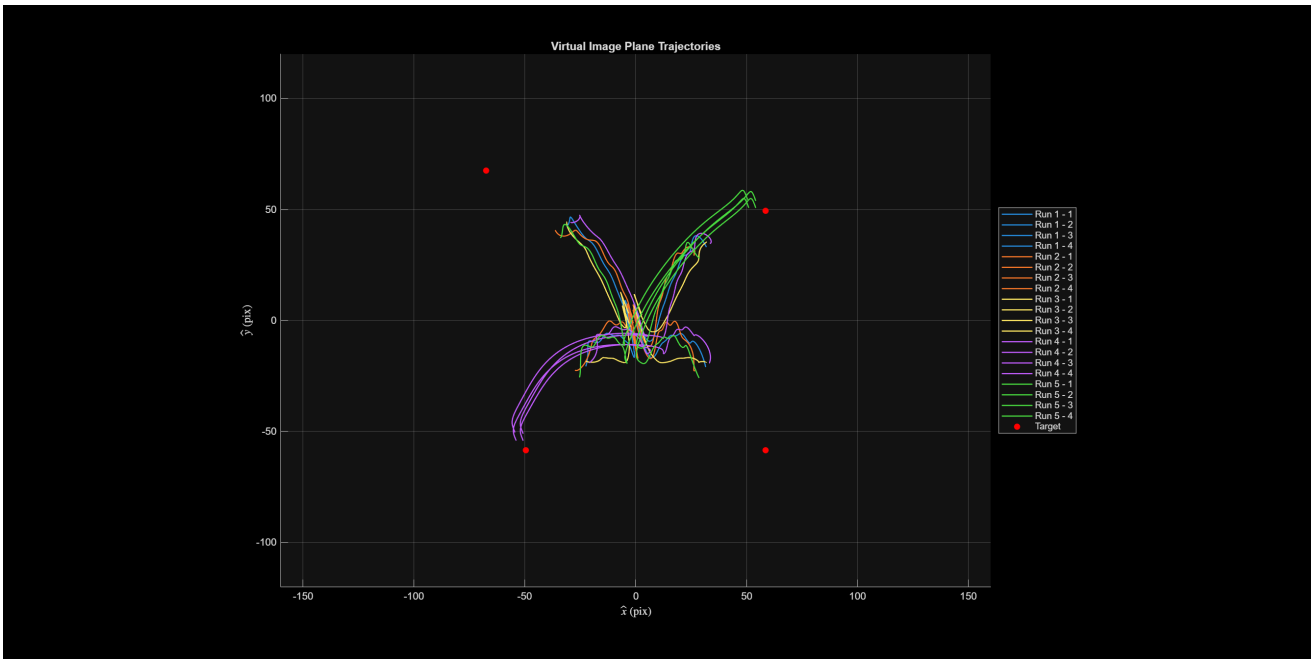


Fig. 9: This graph shows the evolution of virtual image points $\hat{\boldsymbol{r}} = [\hat{x}, \hat{y}]^{\top}$ in the virtual image plane for different initial conditions (*Table* II) in idealistic conditions, when the target is undergoing lissajous motion on a horizontal plane.

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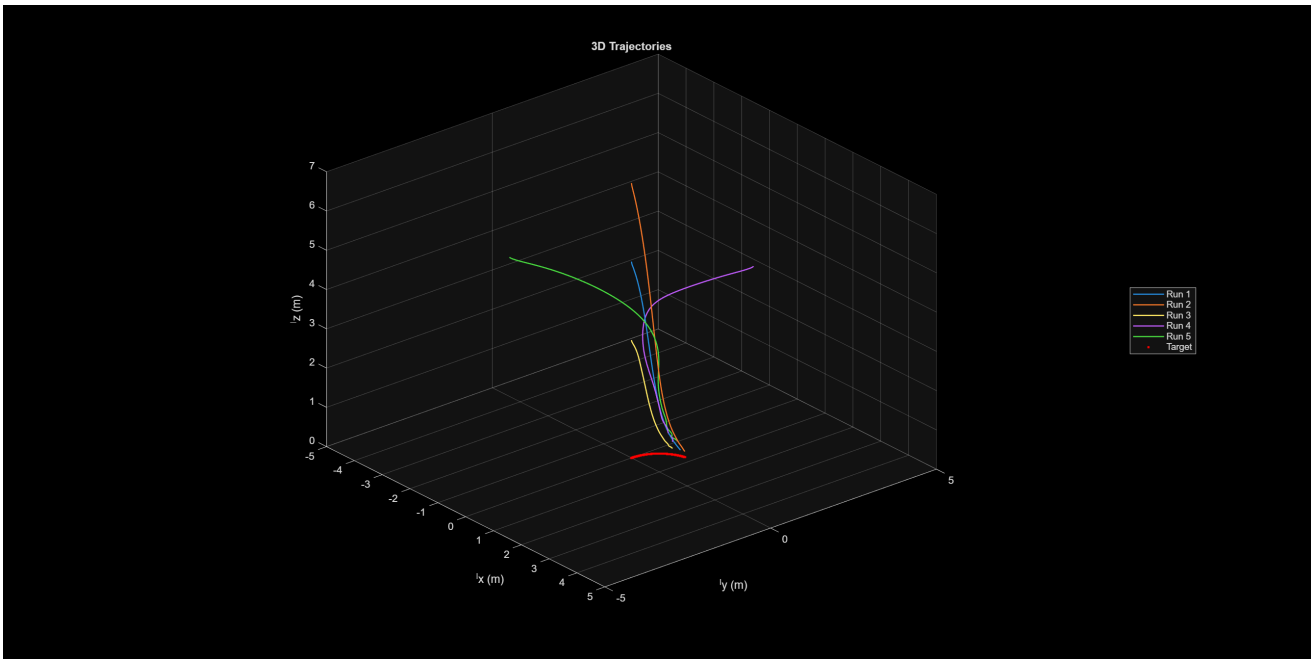


Fig. 10: This 3-D plot shows the evolution of UAV position in inertial frame ${}^{\mathcal{I}}\boldsymbol{r} = [{}^{\mathcal{I}}x, {}^{\mathcal{I}}y, {}^{\mathcal{I}}z]^\top$ for different initial conditions (*Table* II) in idealistic conditions, when the target is undergoing circular motion on a horizontal plane.

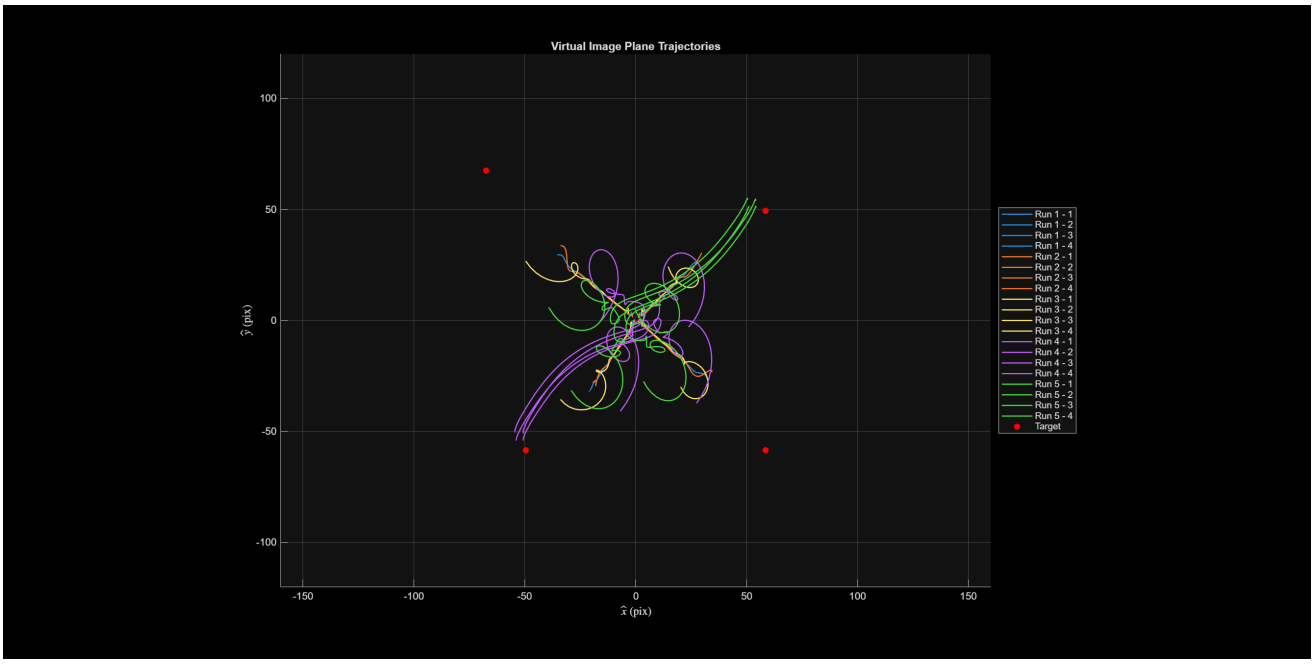


Fig. 11: This graph shows the evolution of virtual image points $\hat{\boldsymbol{r}} = [\hat{x}, \hat{y}]^\top$ in the virtual image plane for different initial conditions (*Table* II) in idealistic conditions, when the target is undergoing circular motion on a horizontal plane.

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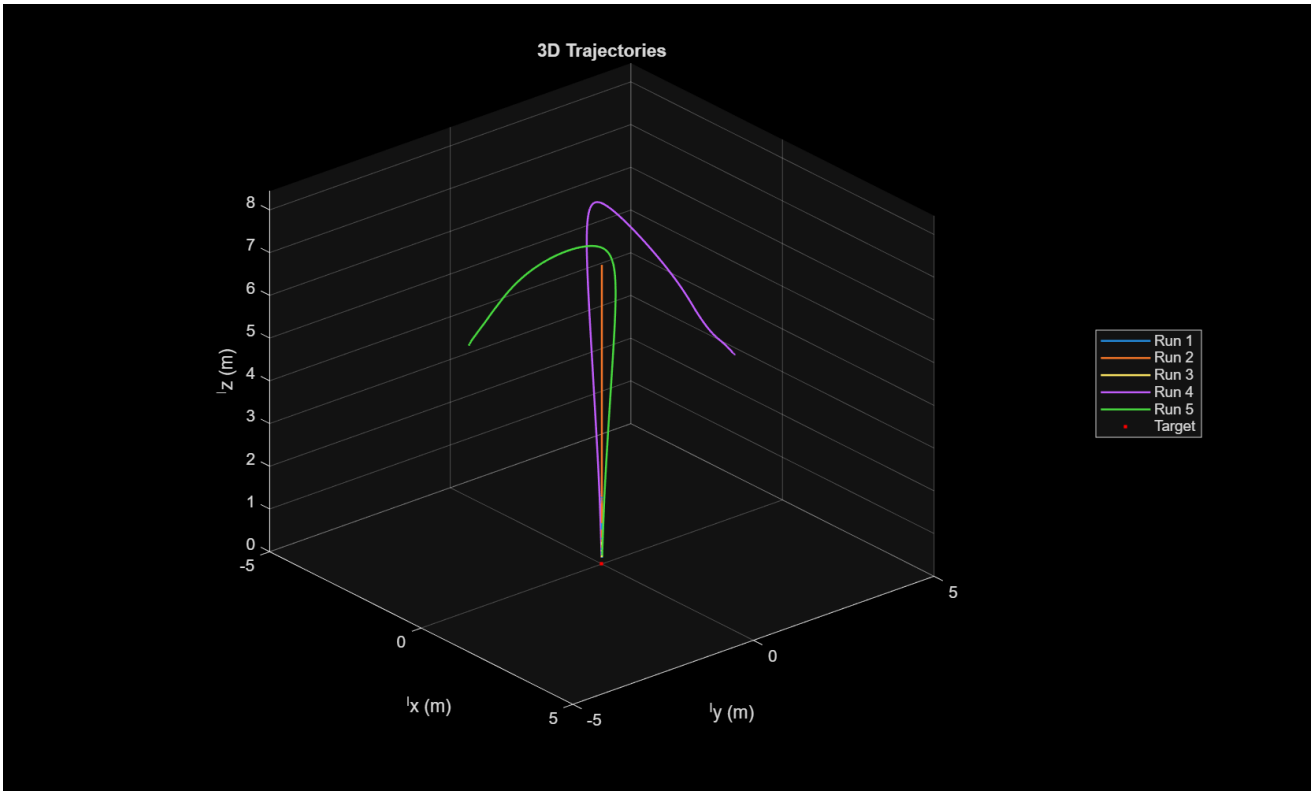


Fig. 12: This 3-D plot shows the evolution of UAV position in inertial frame ${}^{\mathcal{I}}\boldsymbol{r} = [{}^{\mathcal{I}}x, {}^{\mathcal{I}}y, {}^{\mathcal{I}}z]^{\top}$ for different initial conditions (*Table* II) in the presence of model uncertainties and exogeneous perturbations, when the target is static on a horizontal plane.

partment of Mechanical Engineering and the Robert Bosch Centre for Cyber-Physical Systems at Indian Institute of Science, Bengaluru, Karnataka, India. His research interests are broadly in the areas of dynamical systems theory, data-driven nonlinear dynamics and control, and autonomous vision.

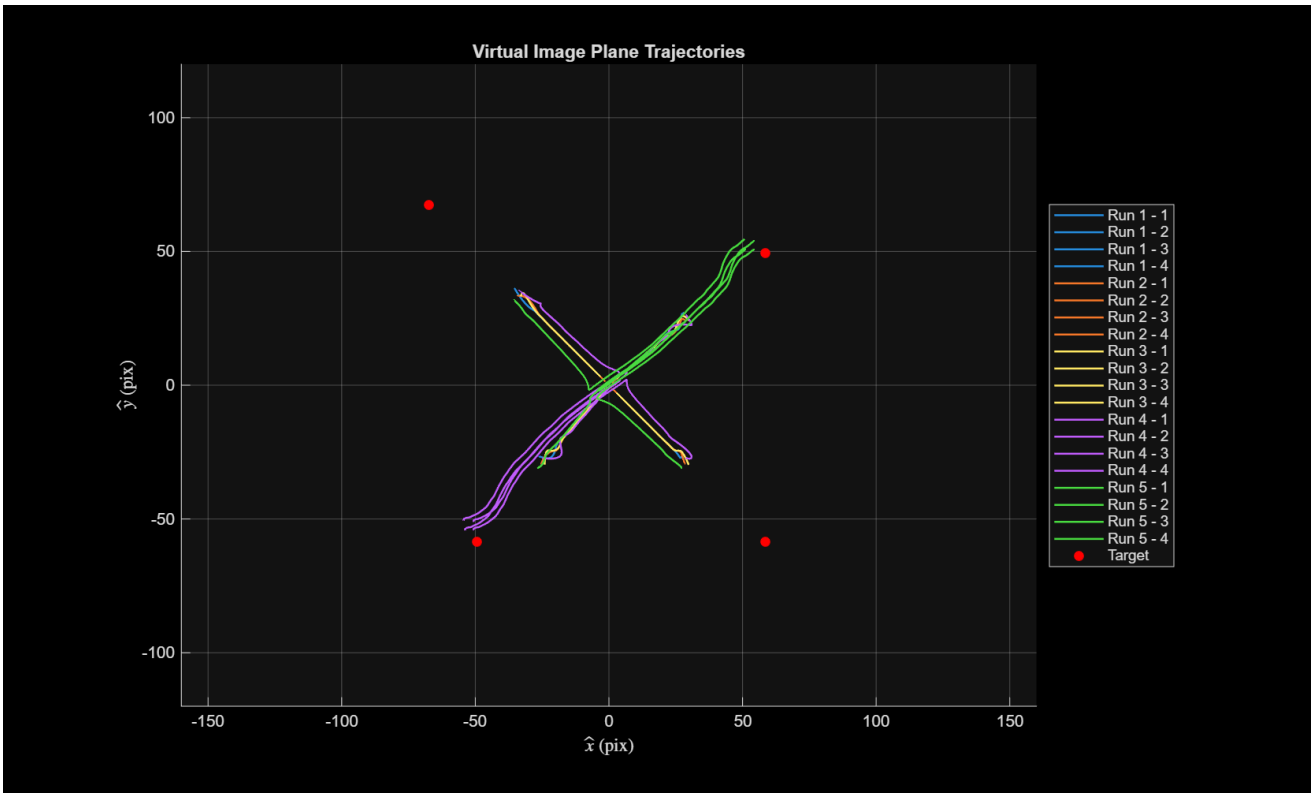


Fig. 13: This graph shows the evolution of virtual image points $\hat{\boldsymbol{r}} = [\hat{x}, \hat{y}]^\top$ in the virtual image plane for different initial conditions (*Table* II) in the presence of model uncertainties and exogeneous perturbations, when the target is static on a horizontal plane.

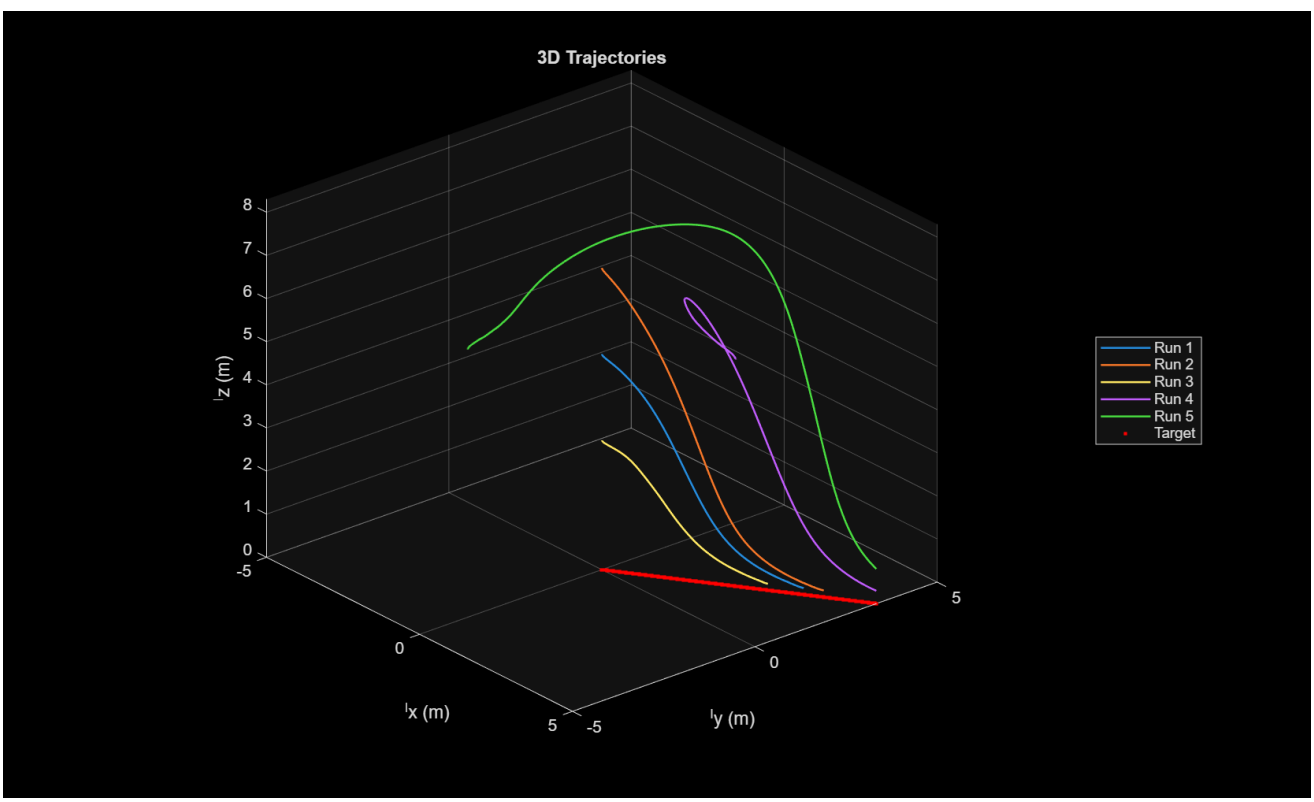


Fig. 14: This 3-D plot shows the evolution of UAV position in inertial frame ${}^{\mathcal{I}}\boldsymbol{r} = [{}^{\mathcal{I}}x, {}^{\mathcal{I}}y, {}^{\mathcal{I}}z]^\top$ for different initial conditions (*Table* II) in the presence of model uncertainties and exogeneous perturbations, when the target is undergoing linear motion on a horizontal plane.

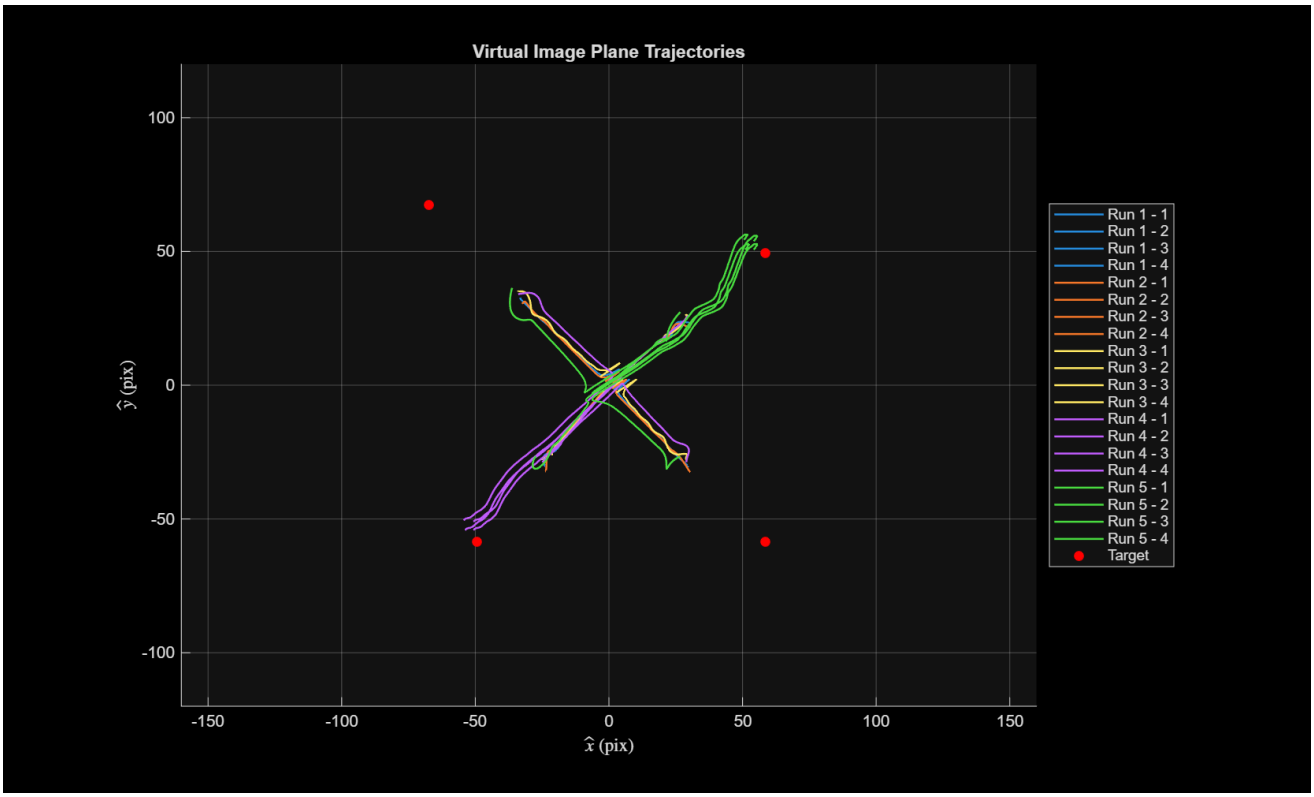


Fig. 15: This graph shows the evolution of virtual image points $\hat{\boldsymbol{r}} = [\hat{x}, \hat{y}]^\top$ in the virtual image plane for different initial conditions (*Table* II) in the presence of model uncertainties and exogeneous perturbations, when the target is undergoing linear motion on a horizontal plane.

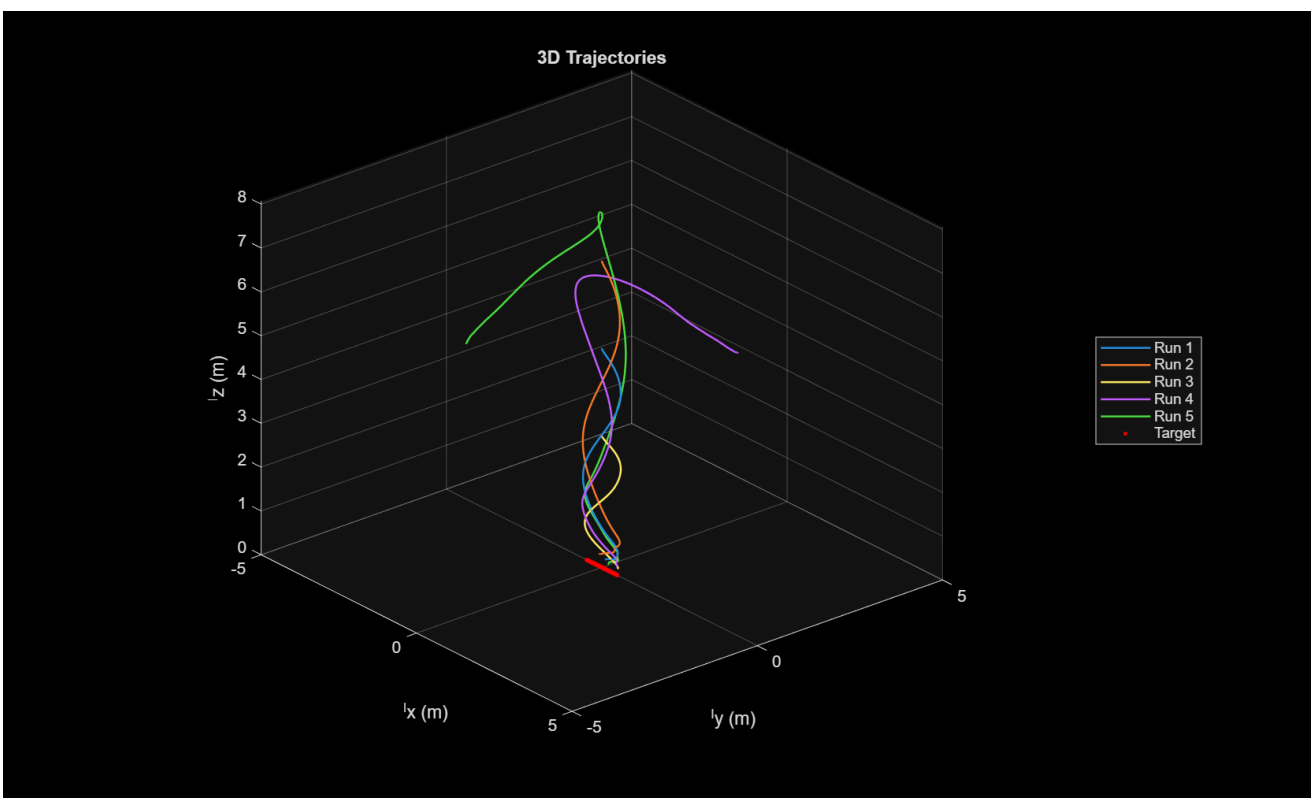


Fig. 16: This 3-D plot shows the evolution of UAV position in inertial frame ${}^{\mathcal{I}}\boldsymbol{r} = [{}^{\mathcal{I}}x, {}^{\mathcal{I}}y, {}^{\mathcal{I}}z]^\top$ for different initial conditions (*Table* II) in the presence of model uncertainties and exogeneous perturbations, when the target is undergoing sinusoidal motion on a horizontal plane.

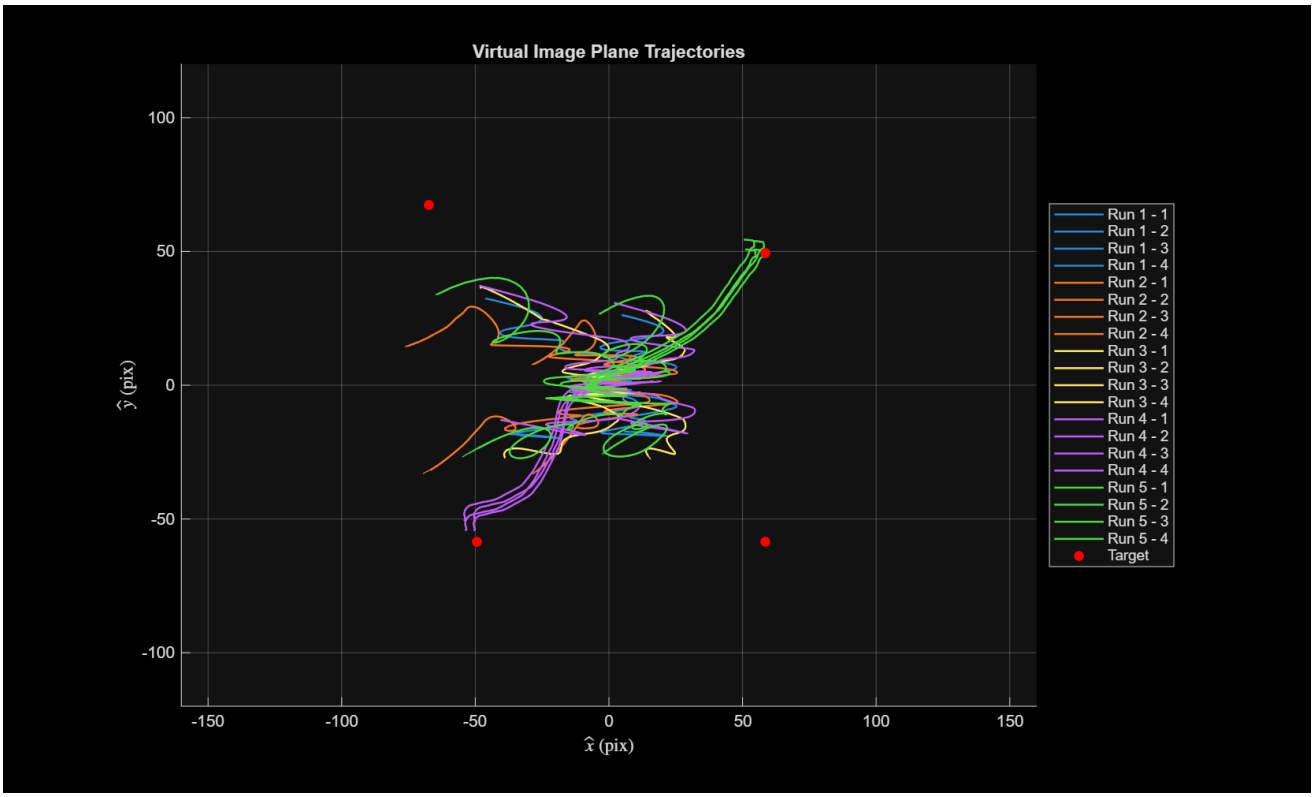


Fig. 17: This graph shows the evolution of virtual image points $\hat{\boldsymbol{r}} = [\hat{x}, \hat{y}]^\top$ in the virtual image plane for different initial conditions (*Table* II) in the presence of model uncertainties and exogeneous perturbations, when the target is undergoing sinusoidal motion on a horizontal plane.

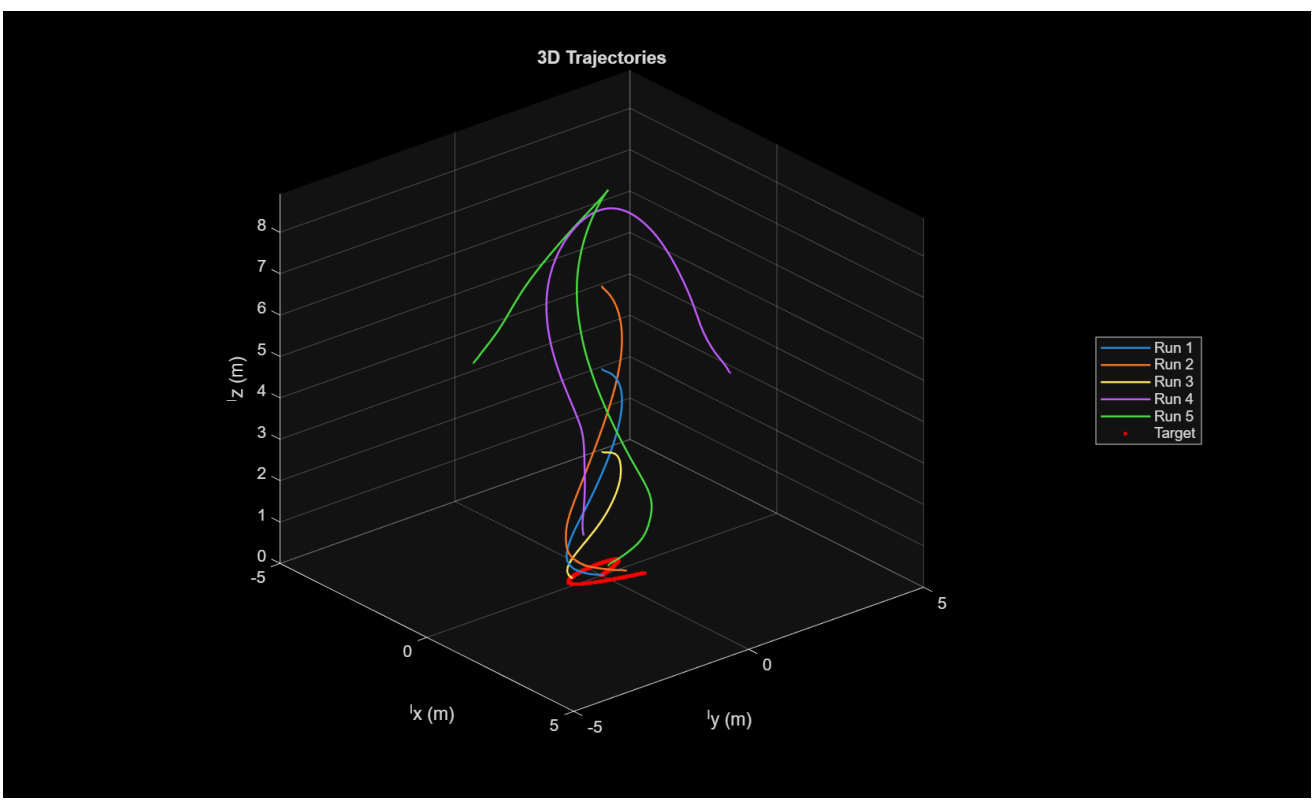


Fig. 18: This 3-D plot shows the evolution of UAV position in inertial frame ${}^{\mathcal{I}}\boldsymbol{r} = [{}^{\mathcal{I}}x, {}^{\mathcal{I}}y, {}^{\mathcal{I}}z]^\top$ for different initial conditions (*Table* II) in the presence of model uncertainties and exogeneous perturbations, when the target is undergoing lissajous motion on a horizontal plane.

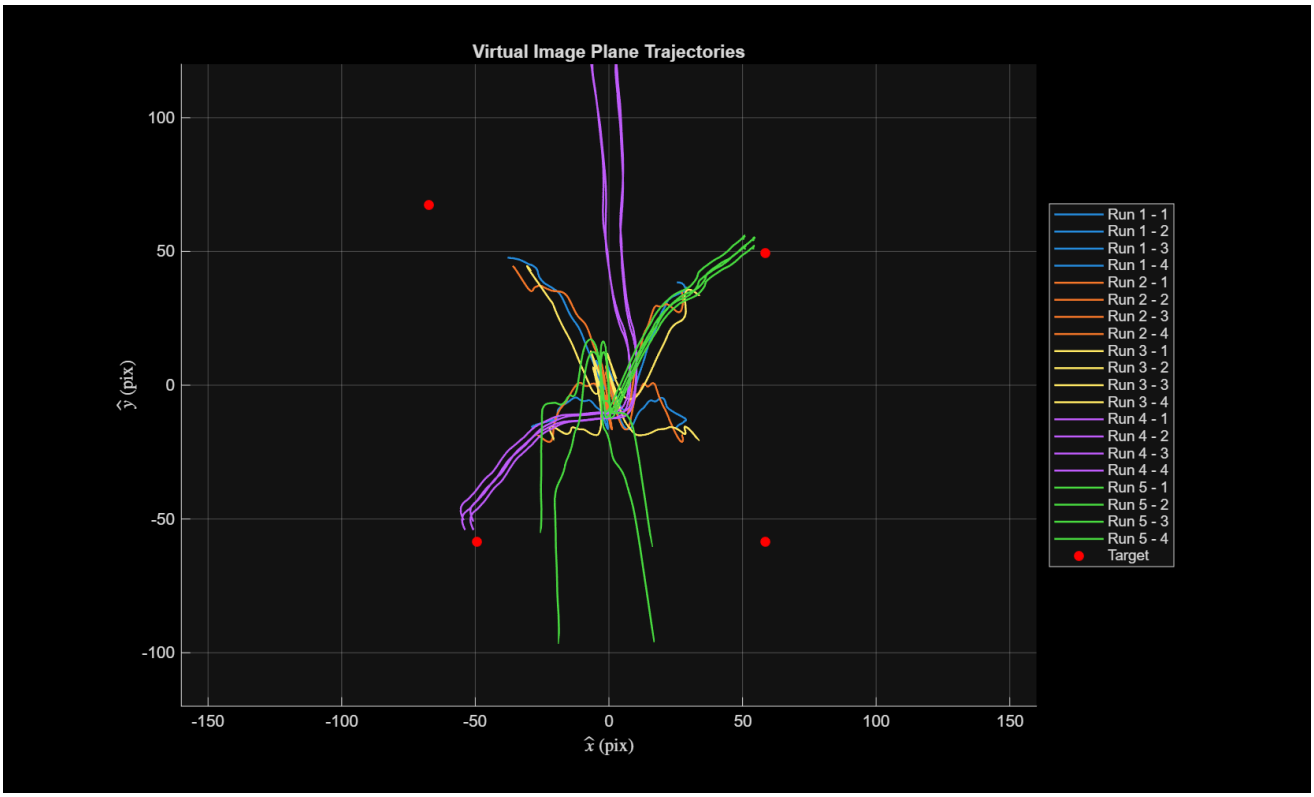


Fig. 19: This graph shows the evolution of virtual image points $\hat{\boldsymbol{r}} = [\hat{x}, \hat{y}]^\top$ in the virtual image plane for different initial conditions (*Table* II) in the presence of model uncertainties and exogeneous perturbations, when the target is undergoing lissajous motion on a horizontal plane.

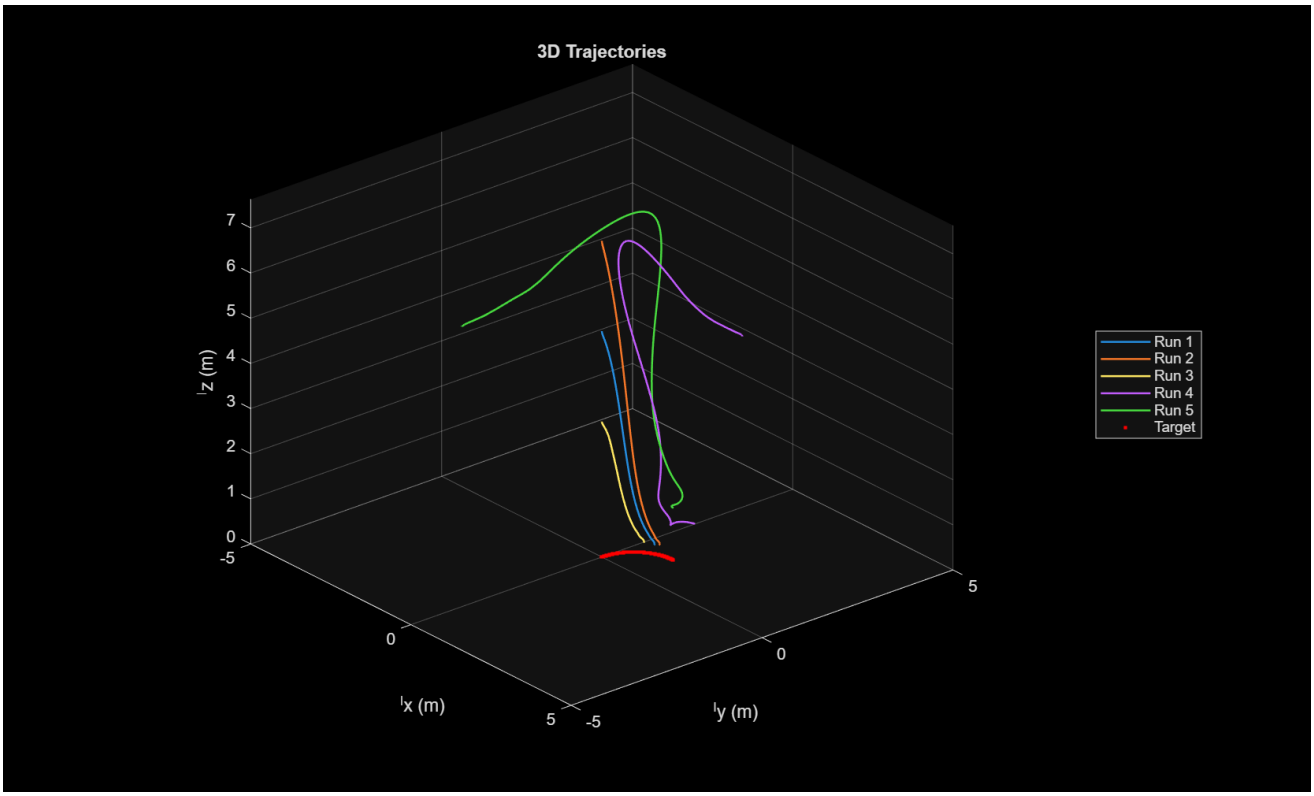


Fig. 20: This 3-D plot shows the evolution of UAV position in inertial frame ${}^{\mathcal{I}}\boldsymbol{r} = [{}^{\mathcal{I}}x, {}^{\mathcal{I}}y, {}^{\mathcal{I}}z]^\top$ for different initial conditions (*Table* II) in the presence of model uncertainties and exogeneous perturbations, when the target is undergoing circular motion on a horizontal plane.

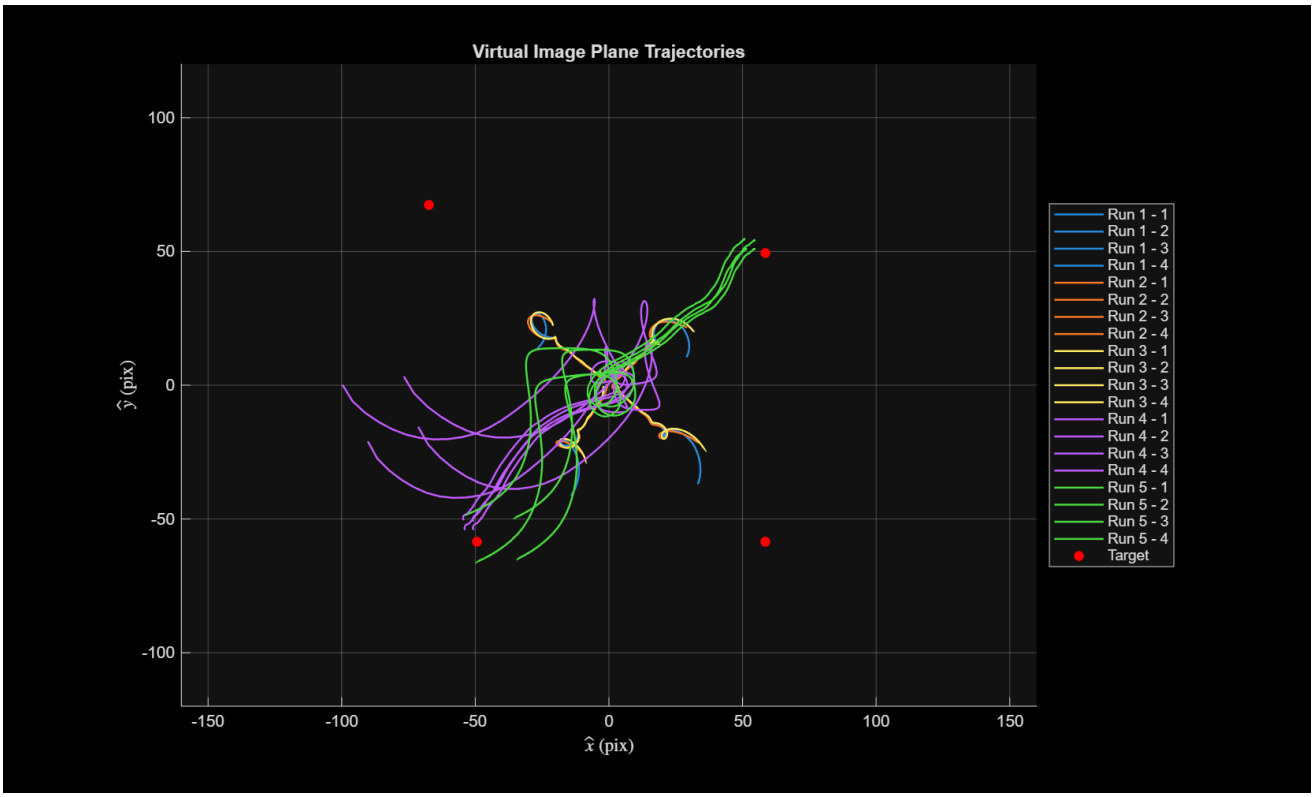


Fig. 21: This graph shows the evolution of virtual image points $\hat{\boldsymbol{r}} = [\hat{x}, \hat{y}]^\top$ in the virtual image plane for different initial conditions (*Table* II) in the presence of model uncertainties and exogeneous perturbations, when the target is undergoing circular motion on a horizontal plane.